The Deposit Channel of Quantitative Easing

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Abstract

Quantitative easing (QE) creates bank deposits when the Federal Reserve buys securities from non-banks, and this new money enters the banking system at a few primary dealers. I measure each bank's and county's exposure to these points of entry and show that it predicts deposit growth when the Fed is buying but not between programs. Controlling for asset-side QE exposure, deposit-channel exposure predicts additional lending and reaches far more banks. Within banks, lending rises more in exposed counties, favoring mortgages over small-business loans. Exposure also raises non-tradable employment and house prices in both QE episodes. Where money is created matters.

Keywords: Quantitative easing, deposit channel, monetary policy transmission, bank lending, regional heterogeneity.

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1 Introduction

The Federal Reserve ("the Fed") bought more than $8 trillion of long-term Treasuries and mortgage-backed securities (MBS) in two rounds of Quantitative Easing (QE), the first following the Global Financial Crisis (GFC) and the second following the Pandemic. The textbook description of QE is an asset swap: the Fed buys a bond from a bank and pays with reserves, so the bank's liabilities are unchanged and only the composition of its assets shifts. Most of the bonds the Fed bought, however, came from outside the banking system (Carpenter et al., 2015). When a non-bank sells a bond to the Fed, it cannot be paid in reserves. It is paid in bank deposits. QE therefore created new deposits on a macroeconomic scale, and this paper shows that these deposits had large real effects — on bank lending, local employment, and house prices — that were concentrated in the places where the new money entered the banking system.[1]

[Insert Figure 1 Here]

Deposits created by QE do not arrive everywhere at once. Every Fed purchase is executed through one of roughly two dozen Primary Dealers (PDs). The seller is paid by the dealer, so the new deposit is first credited at the dealer's bank. From there the money moves outward to the rest of the banking system. The new money created by QE thus propagates across banks gradually, with its largest immediate effects at banks located near the PD banks.[2] I call the transmission of QE through this liability side of bank balance sheets the deposit channel of QE.

The deposit channel differs from the asset-swap channel in two ways. First, it works on the quantity of bank funding rather than the composition of bank assets. A bank need not hold any eligible securities, or report any reserves, to be affected; it is exposed simply by taking in QE-generated deposits. The channel therefore reaches the whole

[1] The textbook asset-swap and the deposit channels are illustrated in Figure 1.

[2] This is an instance of the Cantillon Effect (Cantillon, 1931), which holds that money creation has non-neutral effects because new money does not reach all agents at the same time, as the metaphorical "helicopter drop" of money would (Friedman, 1969; Bernanke, 2002).

universe of deposit-taking banks, including the many small institutions that hold few securities and do not report reserves, and that are invisible to the asset- and reserve-based exposure measures used in prior work (Kandrac and Schlusche, 2021; Acharya et al., 2024).[3] Second, the channel has a geography, for two reasons. The new money first enters the banking system at a small number of institutions, and local banking markets, and the banks operating in them, overlap with these institutions' networks to very different degrees, so the money is likely to reach some markets sooner and in larger amounts than others. A national policy is thereby transmitted into an uneven distribution of liquidity across local banking markets.

To measure exposure to the deposit channel I construct an instrument from the Fed's own purchase records, which are publicly disclosed at the transaction level. The instrument follows the money from its point of entry outward in two steps. First, the Fed's net purchases from each of the eight U.S. primary dealers are allocated across counties according to the dealer's 2008 branch and office footprint.[4] Second, within each county, the resulting inflow is allocated across the banks operating there in proportion to each bank's 2008 share of county deposits. These allocations are predicted flows, not the actual flows into each county and bank, which I do not observe. I use them to predict the actual flows, since the money is likely to reach first the markets in which the dealer's bank operates and serves its customers. All weights are fixed at their 2008 values, before QE began.[5]

The instrument does not require a particular account of how deposits travel from the dealers to other banks. It requires only that the money created by QE enters the banking

[3]Since 2017, banks with only domestic offices and assets under $5 billion may file the simplified Form 051, which omits reserve balances, and banks with assets under $300 million filing Form 041 are not required to report them. Over 2009–2023, more than 70% of bank-quarter observations lack reserve information.

[4]The eight U.S. primary dealers are Bank of America, Morgan Stanley, Goldman Sachs, Citigroup, JPMorgan Chase, Wells Fargo, Jefferies, and Cantor Fitzgerald. They accounted for 53% and 57% of the Fed's net purchases of Treasuries and MBS, respectively. I restrict attention to domestic dealers because their U.S. networks can be located; foreign dealers operate through a few branches and agencies whose geography carries little information.

[5]There was no market-wide anticipation of QE in mid-2008: the federal funds rate stood at 2% and the Fed publicly prioritized inflation risk (Bernanke, 2008). Large-scale asset purchases were announced in late November 2008.

system at a small set of institutions that are unevenly distributed across the country, which follows from how the Fed conducts its purchases, and that markets with greater overlap with those institutions receive more of the money. The second requirement is what the first stage tests. If the location of money creation did not matter, a market's overlap with the dealer networks would not predict its deposit growth. It does. The instrument is strongly relevant at the county, bank-county, and bank levels.

Nearly all of the variation in the instrument comes from the aggregate volume of Fed purchases interacted with each market's 2008 overlap with the dealer networks. The identifying assumption is therefore that, conditional on the fixed effects and controls, markets with greater overlap would not have had different credit, employment, or house-price outcomes during heavy-purchase periods for reasons other than the deposits QE created there. The main concern is that high-overlap markets respond differently to the other channels through which QE works at the same time, such as lower mortgage rates and portfolio rebalancing. I address this concern in four ways. First, at the bank-county level, I compare the same bank's lending across the counties it serves in the same year. Bank-by-year fixed effects absorb anything specific to a bank in a given year, including its holdings of MBS and Treasuries, its exposure to duration risk, and its desire to rebalance its portfolio after the Fed's purchases, so differences in asset composition across banks cannot explain the within-bank results. County-by-year fixed effects absorb local credit demand and any county-wide response to lower mortgage rates. Second, the bank-level and county-level regressions cannot include these fixed effects, so there I control directly for exposure to the Fed's purchases through banks' MBS and Treasury holdings. Third, overlap predicts deposit growth only while the Fed is buying, and not in the years between QE programs. A persistent difference between high- and low-overlap markets would not switch on and off with the Fed's purchases. Fourth, the composition of the responses fits a shock to local bank credit. The mortgage response is strongest for jumbo loans, which fall outside the conforming market whose rates the Fed's MBS purchases lowered most directly (Di Maggio, Kermani and Palmer, 2020), and employment rises in non-tradable industries, which depend on local demand,

but not in tradable industries, which do not.

A deposit inflow need not expand lending. Deposits and lending are linked in both directions: deposits fund loans, and a new loan also creates a deposit, at least initially, in the borrower's account (McLeay, Radia and Thomas, 2014). What ultimately limits a bank's lending is how far it can expand its balance sheet, which depends on its capital. A bank that receives a QE-generated deposit also receives the reserves that settle it.[6] The new funding and reserves make it easier for the bank to meet payment outflows and allow it to lend more for a given tolerance for liquidity risk (Bianchi and Bigio, 2022). But the inflow also enlarges the bank's balance sheet without adding equity, which uses up capacity under its capital and leverage requirements and can hold back lending at banks for which those requirements bind (Diamond, Jiang and Ma, 2024). Which effect is larger is an empirical question. In my sample, which consists largely of smaller banks, the first effect dominates, and QE-generated deposits raise lending.[7]

The new lending also need not stay in the markets where the deposits arrive. Banks can move funds across their branch networks, and Gilje, Loutskina and Strahan (2016) show that they lend part of a local deposit windfall in other markets. Using the within-bank comparison described above, I find that lending rises more in the counties where the bank's exposure is larger.[8] This comparison does not tell us what fraction of the new funding is lent locally. A bank can lend relatively more in its high-exposure counties

[6]Because interbank payments are settled in reserves, a QE-generated deposit cannot arrive at a bank without the reserves that settle it. When the Fed pays for a bond, it credits the reserve account of the seller's bank (or of its correspondent), and when the deposit later moves to another bank, reserves move with it. An exposure measure that predicts deposit inflows must therefore also predict reserve inflows at the banks that report them, and that it does is not evidence of a separate reserve channel: the deposit and the reserve are the liability and asset sides of the same inflow. The distinction that matters for this paper is instead between reserves that arrive with new deposits, which expand the bank's balance sheet, and reserves that a bank obtains by selling its own securities to the Fed, which only change its composition. The controls for each bank's MBS and Treasury holdings capture the latter.

[7]I exclude banks that are subsidiaries of primary dealers throughout, so that the estimates reflect deposits reaching other banks as the money spreads outward, not the dealers' own affiliates or their internal capital markets. Because the primary dealers include several of the largest U.S. bank holding companies, the banks in my sample are typically smaller institutions.

[8]How much of a local deposit inflow a bank lends locally depends on how well its internal capital market works. If funds moved freely within the bank, the new funding would go to the bank's best lending opportunities wherever the deposits arrived. Information and agency frictions between headquarters and local managers can instead tie lending to local funding (Stein, 1997; Scharfstein and Stein, 2000). The within-bank estimates therefore reflect both the arrival of the new deposits and the frictions in banks' internal capital markets, which I do not separate.

and still move funds across its network, and any increase that is common to all of the bank's counties is absorbed by the bank-by-year fixed effects and appears instead in the bank-level estimates. Because banks lend more where more of the money arrives, the uneven entry of QE money translates into uneven local effects on employment and house prices.

I establish four sets of results. First, at the bank level, deposit-channel exposure predicts lending growth over and above the asset-side QE measures used in prior work. Controlling for each bank's exposure to the Fed's MBS and Treasury purchases through its own securities holdings, a dollar of deposit-channel exposure is associated with about $0.28 of additional lending. This is a reduced-form response per dollar of Fed purchases allocated to the bank by the instrument, not an instrumented effect per dollar of realized deposit inflow. The response appears first in commercial lending and builds into retained mortgages more gradually, and banks fund part of the expansion by running down their securities holdings. Second, within banks and across counties, deposit-channel exposure raises local mortgage originations by about three times as much as local small-business lending per dollar of exposure. Third, the channel has real local effects. Across metropolitan counties, and controlling for the counties' exposure to the Fed's Treasury and MBS purchases, a one-standard-deviation increase in deposit-channel exposure raises non-tradable employment cumulatively over eight quarters by 0.5 percent in QE I–III and by 2.7 percent in the pandemic episode. Tradable employment does not respond in either episode, consistent with a local demand shock rather than a shift in local productive capacity. Fourth, the channel raises local house prices. A one-standard-deviation increase in exposure raises the FHFA house price index over two years by 1.9 percentage points in QE I–III and by 1.1 percentage points in the pandemic episode, and quarterly Zillow indices show the same pattern. Exposure also raises home-purchase originations, approval rates, and the number and volume of mortgages banks retain. The effect is strongest for jumbo loans, which banks generally hold on balance sheet rather than sell to the GSEs and which therefore depend more directly on bank balance-sheet capacity.

These effects are present in both QE rounds. The post-2020 housing boom is often associated with pandemic-era QE, but the deposit channel also raised house prices during QE I–III, when the Fed bought securities under very different macroeconomic conditions. The geographic non-neutrality of QE is therefore not specific to the pandemic; it is a recurring feature of how central-bank money creation transmits through the banking system.

My paper contributes to several strands of the literature on QE and bank funding.

First, the results speak to the literature on how QE affects bank lending. Existing studies measure a bank's exposure from the asset side, through the MBS it holds (Rodnyansky and Darmouni, 2017; Chakraborty, Goldstein and MacKinlay, 2020), or from the reserves it accumulates (Kandrac and Schlusche, 2021; Diamond, Jiang and Ma, 2024; Acharya et al., 2024), and they reach different conclusions about the sign of the reserve effect. I extend these results by looking at the liability side of bank balance sheets. When the Fed buys bonds from non-banks, it creates deposits that enter the banking system at a few institutions and then spread to other banks. Exposure to these deposits predicts bank lending over and above the asset-side measures, and it reaches a much broader set of banks, including the many small banks that hold few eligible securities and do not report reserves.

Second, the paper complements Drechsler, Savov and Schnabl (2017), who introduce the deposit channel of conventional monetary policy. In their channel, rising rates let banks with market power widen deposit spreads, deposits flow out, and lending contracts; the channel works on the price of deposits. The deposit channel of QE works on the quantity of deposits and in the opposite direction: it injects new deposits whose incidence depends on a market's overlap with the points of entry, and it expands lending. Deposit quantities, and where they are created, carry first-order importance for the transmission of unconventional policy.

Third, the paper adds to the literature on the regional transmission of monetary policy, which documents that a uniform national policy has uneven local effects (e.g.,

Beraja et al., 2018). That literature attributes the uneven effects to differences across the places that receive the shock, such as the regional distribution of housing equity. I show that the unevenness can also come from the shock itself. QE money is created at a small number of institutions that are unevenly distributed across the country, and its effects on credit, employment, and house prices are largest in the markets that overlap most with those institutions. This is an instance of the Cantillon effect (Cantillon, 1931). To my knowledge, this is the first paper to link QE's local real effects to where the new deposits are created.

Fourth, the paper connects to work on QE and housing. Di Maggio, Kermani and Palmer (2020) show that the Fed's MBS purchases lowered conforming mortgage rates and spurred refinancing and equity extraction; Boddin et al. (2026) show that European QE pushed bondholders' wealth into investment properties; Favara and Imbs (2015) and Berg et al. (2026) show that credit-supply expansions raise house prices. The deposit channel is the bank-side counterpart to these demand-side mechanisms. It expands the liability side of bank balance sheets and thereby the local supply of mortgage credit, including the jumbo loans that banks typically retain, and the resulting house-price gains rebuild the equity that refinancing requires. It also reaches employment through small-business credit and local demand (Mian and Sufi, 2014; Luck and Zimmermann, 2020).

Finally, the paper departs from the high-frequency identification used in much of the QE literature (Gagnon et al., 2011; Krishnamurthy et al., 2011; Krishnamurthy and Vissing-Jorgensen, 2013; Aastveit and Anundsen, 2022), surveyed in Kuttner (2018) and Bernanke (2019), which infers transmission from market reactions in narrow announcement windows. Because the real effects of QE take time to build through financial institutions, my exposure measure is anchored instead on where and when the Fed's purchases actually create deposits, which maps naturally to long-horizon outcomes such as credit, house prices, and employment.

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 develops the instrument and establishes its relevance and identifying assumptions.

Section 4 presents the bank-level and bank-county-level lending results. Sections 5 and 6 present the effects on local employment and house prices, and Section 7 examines the credit-supply mechanism through mortgage approval, retention, and the conforming–jumbo distinction. Section 8 concludes.

2 Data

This section details the data sources used in this paper. The deposit channel instrument is constructed by merging Federal Reserve Historical Transaction Data with the Federal Deposit Insurance Corporation (FDIC) Summary of Deposits (SOD) Data. To assess housing market outcomes, I use loan-level data from the Home Mortgage Disclosure Act (HMDA) alongside market-wide indicators from the Federal Housing Finance Agency (FHFA) and Zillow. I also use Community Reinvestment Act (CRA) data, which provide bank-county small business lending information, to study local credit allocation. These datasets are supplemented with bank-level financial information from the Federal Financial Institutions Examination Council (FFIEC) Call Reports, while local employment outcomes are sourced from the Quarterly Census of Employment and Wages (QCEW). Broader macroeconomic trends are tracked using Federal Reserve Economic Data (FRED). The primary sample spans 2009Q1 to 2023Q4, encompassing multiple rounds of QE. Detailed variable definitions and comprehensive summary statistics are provided in Appendix Tables A.1 and A.2, respectively.

2.1 Federal Reserve Historical Transaction Data

To construct the time series of liquidity injections for each primary dealer, I use historical transaction-level data provided by the Federal Reserve Bank of New York. This dataset records all Federal Reserve security trades with primary dealers — the institutions authorized to transact directly with the Fed in its open market operations — providing granular detail on daily transactions with specific counterparties. Table 1 lists the 26 primary dealers authorized as trading counterparties for the Federal Reserve Bank of New York as of February 2026. The transactions are released with a mandatory two-year disclosure lag; my sample uses the data available at the time of writing, which

extend through 2023Q4. The data encompass a broad range of instruments, including U.S. Treasuries as well as agency residential mortgage-backed securities (MBS) and commercial mortgage-backed securities (CMBS). Each entry identifies the trade and settlement dates, the transaction type (purchase or sale), the par and dollar amounts, the security CUSIP, and the counterparty identity.

[Insert Table 1 Here]

2.2 FDIC SOD Data

To construct the cross-sectional exposure of individual banks to QE, I use the FDIC SOD data, which provide granular, branch-level information on deposits and geographic identifiers for all FDIC-insured institutions. These data are released annually with a reporting date of June 30, and I use the records to compute two key components of the instrument: first, the geographic distribution of each primary dealer across counties, and second, the deposit market share of each bank within those counties. This approach enables a precise mapping of liquidity transmission from primary dealers to local markets and, ultimately, to individual banks.

2.3 FHFA House Price Index (HPI) Data

To measure local house price appreciation, I utilize the FHFA House Price Index (HPI) data, which provides annual indices for all U.S. counties. The HPI is a repeat-sales index that tracks price changes by observing the same property over multiple transaction points. Because residential properties are sold infrequently, the index incorporates appraisals to provide a more continuous evaluation of market values. Although the FHFA offers indices calculated using various fixed base years (e.g., 1990 or 2000), I select the index based on the first available year for each county to maximize longitudinal coverage. This index is constructed using sales prices from mortgages purchased or guaranteed by Fannie Mae and Freddie Mac. Although these conforming mortgages represent the primary source of residential financing, the dataset does not fully capture price dynamics in the jumbo or subprime mortgage sectors, which fall

outside conforming loan limits.

2.4 Zillow Home Value Index (ZHVI) Data

To supplement the FHFA HPI, I utilize the Zillow Home Value Index (ZHVI), which provides monthly home value data at the county level. The ZHVI is designed to reflect the value of homes within the 35th to 65th percentile range and is available in a seasonally adjusted format. In contrast to the repeat-sales methodology used by the FHFA, Zillow employs a hedonic machine-learning model — the "Zestimate" — to estimate the current market value of every home within a county, regardless of whether a recent transaction has occurred. A primary advantage of the ZHVI is its broader market coverage, as it incorporates all transaction types, including cash sales and jumbo mortgages that are excluded from the FHFA's conforming loan sample. For this study, I aggregate the monthly indices to a quarterly frequency by calculating the arithmetic mean of the monthly observations within each quarter.

2.5 HMDA Data

To complement the FHFA and Zillow indices and provide a comprehensive view of the local housing market, I utilize HMDA data. As the primary source for evaluating the credit supply channel, HMDA offers granular, loan-level records for the near-universe of U.S. mortgage applications. For each application, the dataset identifies the originating institution and provides detailed loan characteristics, including geographic location, loan purpose (purchase versus refinance), and loan amount. Crucially, it allows me to distinguish between the originate-to-hold model — where the lender retains the loan on its balance sheet — and the originate-to-distribute model, where the loan is sold to GSEs or private entities. Beginning in 2018, HMDA reporting expanded to include several critical risk indicators. These include a flag for whether a loan falls within the FHFA Conforming Loan Limit (CLL) — enabling the precise classification of conforming versus jumbo loans — as well as property values, loan-to-value (LTV) ratios, and debt-to-income (DTI) ratios. Furthermore, the data contain demographic information such as the race, ethnicity, and sex of the applicant, facilitating an analysis

of credit distribution across various borrower segments.

2.6 CRA Data

To study local credit allocation across loan categories, I use CRA data, which report the annual volume of small business lending by bank and county. Lenders subject to CRA reporting disclose, for each county in which they operate, the number and dollar amount of small business loans originated, including a separate tabulation of loans to businesses with gross annual revenue below $1 million. I aggregate these records to the bank-county-year level and merge them with the HMDA mortgage data, which allows me to compare the response of small business and mortgage lending to the deposit channel within the same bank-county.

2.7 FFIEC Call Report Data

To evaluate the impact of QE on bank-level financial outcomes, I utilize FFIEC Call Report data accessed via the Wharton Research Data Services (WRDS). This dataset provides comprehensive accounting information derived from the Consolidated Reports of Condition and Income, which all insured commercial banks are required to file. For each bank-quarter, I retain only the most recent submission to ensure the inclusion of amended filings. Because flow variables in the Call Reports — such as net income and interest expense — are reported on a year-to-date (YTD) basis, I transform these measures into discrete quarterly flows by subtracting the value from the preceding quarter within the same fiscal year. This adjustment ensures a consistent panel of bank performance at a quarterly frequency.

2.8 QCEW Data

To study the effect of QE on local labor markets, I use QCEW data, which provide county-level employment by industry at both quarterly and annual frequencies. I use the quarterly data, aggregating across industries to distinguish employment in the tradable and non-tradable sectors, following the classification of Mian and Sufi (2014).

2.9 FRED Data

I supplement the analysis with aggregate macroeconomic time series from FRED to account for the broader economic environment. These variables capture market conditions prevailing during the various phases of QE implementation and absorb systemic trends.

3 Empirical Strategy

This section outlines the empirical strategy used to identify the deposit channel of QE. I provide a step-by-step construction of the instrument at the county, bank-county, and bank levels to trace its granular transmission, and then demonstrate the validity of the instrument.

3.1 Instrument Construction

My empirical strategy is based on the premise that QE-generated deposits do not land evenly across geography. Instead, they concentrate in areas of high financial business intensity, for two reasons. First, the bond sellers that sell securities to the Fed are likely concentrated in major financial hubs, so the banks that hold their transaction accounts are more likely to receive the initially created deposits. Second, within a given area, banks with a larger local business presence are more likely to absorb a larger share of these QE-generated deposits.

I measure the geographic intensity of QE-generated deposit creation using the footprint of primary dealers — the financial institutions authorized to trade directly with the Federal Reserve in its open market operations. Although each deposit is ultimately booked at the seller's bank rather than at the dealer, primary-dealer presence is a natural proxy for where deposits are created, because dealer activity and the institutional investors who sell securities to the Fed are concentrated in the same financial centers. Securities trading, custody, and institutional cash management all cluster in a small number of financial hubs, so the markets where primary dealers are most active are also the markets where the sellers — and the banks that hold their

balances — are most heavily located. Primary-dealer presence therefore captures the local intensity of the institutions whose securities sales create deposits. The validity of this proxy is ultimately an empirical question, which I address directly through the first-stage relevance results in Section 3.2.

This uneven distribution gives rise to two features that are distinctive to the deposit channel. First, banks operating in different areas receive different amounts of QE-generated deposits. Second, and more importantly, even within a single bank, branches in different locations receive different amounts of these deposits, so the deposit channel generates variation within as well as across banks. This within-bank variation is central to my identification strategy, as it allows me to compare lending across branches of the same bank that are differentially exposed to QE-generated deposits.

I start by identifying the net dollar amount of security purchases for each of the eight U.S. primary dealers (pd) in each quarter (t) conducted by the Federal Reserve's open market operations. This amount is intended to proxy for the deposit-creation intensity associated with each dealer. I focus on the eight domestic dealers because they source most of the securities they sell to the Fed from the U.S. market, thus creating domestic deposits that are absorbed by domestic banks, which are required to report their detailed branch-level deposits in the annual SOD data. These eight domestic primary dealers cover more than 1,700 counties — more than half of the total counties in the U.S. — which provides enough geographical variation to capture the full picture of the deposit channel.

I then distribute the net purchase amount of each primary dealer to the counties (c) it covers according to each county's 2008 business share, and aggregate across dealers to obtain a county-level proxy for the deposit channel. I proxy a dealer's business share using its 2008 local deposit share for dealers with a bank affiliate, and its main-office square footage and employee headcount for dealers without one; because the latter dealers' activities are heavily concentrated in New York, this assigns the greatest weight to New York County.[9] I construct these business-share measures at their 2008

[9]Appendix Table A.3 lists the counties covered by dealers with a 2008 bank affiliate, and Appendix

levels because this predates the start of QE, which helps overcome the reverse-causality problem and alleviate anticipation concerns. In this step, I obtain the county-level instrument for the deposit channel:

$$Z_{c,t} = \sum_{pd \in PD_c} \frac{\text{Business}_{pd,c}}{\text{Business}_{pd}} \times \text{Net Purchase}_{pd,t} \tag{1}$$

where PD_c denotes the set of primary dealers with a physical presence in county c.[10]

In the second step, I allocate the county-level deposit creation measure to each bank (b) that operates in that county, according to each bank's business share, which I proxy using the bank's 2008 deposit share for the same reason as above, to obtain the bank-county-level instrument for the deposit channel:

$$Z_{b,c,t} = \frac{\text{Deposits}_{b,c}}{\text{Deposits}_{c}} \times Z_{c,t} \tag{2}$$

where $Z_{c,t}$ is the county-level inflow from Equation (1).

In the final step, I aggregate the bank's deposit inflow measures across all counties (B_c) where it maintains a physical presence to obtain a bank-level measure for the deposit channel:

$$Z_{b,t} = \sum_{c \in B_c} Z_{b,c,t} \tag{3}$$

Throughout the analysis, I exclude banks that are subsidiaries of primary dealers, so that the estimated effects reflect QE-generated deposits flowing to other banks operating in dealer-active markets rather than to the dealers' own banking affiliates. This restriction keeps the results from being confounded by the internal capital markets of the primary-dealer organizations themselves.

The hierarchical construction of this instrument is illustrated in Figure 2. As de-

Table A.4 reports the estimated business shares for the remaining dealers. The key results are robust to using only bank-affiliated dealers or to alternative weighting schemes for the others.

[10] I restrict the sample to the 50 U.S. states and the District of Columbia to ensure data consistency and focus on the integrated domestic banking market.

signed, the instrument captures two distinct dimensions of variation: time-series variation driven by the fluctuating scale of Federal Reserve security purchases over time, and cross-sectional variation arising from the heterogeneous exposure of specific counties and banks to QE-generated deposits. This dual-variation framework allows for the identification of QE's impact across both time and space, reflecting the uneven distribution of QE-generated deposits throughout the financial system.

[Insert Figure 2 Here]

3.2 Instrument Relevance

For the deposit channel measure of QE to serve as a valid instrument, it must satisfy the relevance condition; that is, the instrument must significantly predict deposit changes at the county, bank-county, and bank levels. I report the county-level and bank-county-level first stages, which rely on the annual SOD data, in Appendix Tables A.5 and A.6, and the bank-level first stage, which uses the quarterly Call Report data, in Table 2. The annual frequency of the SOD data makes the county- and bank-county-level coefficients harder to interpret in levels. Over a full year, a dollar of QE-created deposits does not remain static: as banks lend against the inflow, each new loan itself creates a new deposit, and successive rounds of lending can generate cumulative annual deposit growth in a market that exceeds the size of the initial injection. The estimated pass-through in the annual specifications can therefore exceed one. Because this process plays out over time, it is far less pronounced at the quarterly horizon of the Call Report data, which is one reason the quarterly bank-level estimates in Table 2 remain below one. I therefore treat the annual specifications as evidence of relevance and geographic incidence rather than as estimates of a structural pass-through, and present the cleaner quarterly bank-level estimates in the main text.

I begin with the county-level first stage, reported in Appendix Table A.5, based on the following specification:

$$\Delta\text{Deposits}_{c,t}/\text{Deposits}_{c,t-1} = \beta Z_{c,t}/\text{Deposits}_{c,t-1} + \delta_c + \gamma_t + \epsilon_{c,t} \tag{4}$$

where $\Delta\text{Deposits}_{c,t}$ represents the change in total deposits in county c from year $t-1$ to t, and $Z_{c,t}$ is the county-level deposit channel instrument as defined in Equation (1). To mitigate the influence of outliers, both variables are scaled by lagged total county deposits. To account for unobserved heterogeneity, I include county fixed effects (δ_c) and year fixed effects (γ_t). The model is estimated at an annual frequency, consistent with the reporting cycle of the FDIC SOD data. Because SOD records are updated annually as of June 30, I adjust the net purchase flows for each primary dealer to ensure precise temporal alignment. Specifically, in this empirical specification, year t spans the period from July 1 of year $t-1$ to June 30 of year t. This alignment is intended to align the deposit inflows measured in $Z_{c,t}$ with the period in which the deposit changes are recorded. Finally, since the full sample contains only 15 years, two-way clustering at the county and year levels may result in biased standard errors due to the small number of clusters in the time dimension. To address this, I two-way cluster standard errors at the county and state-year levels, providing more robust inference while accounting for both local persistence and state-specific annual shocks.

The county-level results do more than establish relevance; they directly support the economic logic of the instrument's construction. Appendix Figure A.1 illustrates this geographically: the spatial distribution of the county-level instrument closely tracks the spatial distribution of realized county deposit growth, with both concentrated in the same markets. The regression estimates confirm this pattern. The instrument is a strong predictor of county deposit growth during active purchase periods — the coefficient is 1.00 during the QE I–III era (column 2) and 1.68 during the Pandemic QE era (column 3) — but its predictive power collapses during the No-QE years (column 4), where the coefficient falls to 0.27 and is significant only at the ten percent level, with a first-stage F-statistic below four. This contrast is central to the instrument's interpretation. If the instrument merely captured a secular tendency for deposits to accumulate in

large, financial-center markets — for example, through flight-to-safety flows or the trend growth of big banks' deposit franchises — it would predict deposit growth in those markets regardless of whether the Fed was purchasing securities. Instead, the relationship switches on precisely when QE is active and switches off when it is not, and its magnitude scales with the intensity of purchases across episodes. This pattern is difficult to reconcile with any explanation other than QE-driven deposit creation, and it confirms that the instrument captures the geographic incidence of QE-generated deposits rather than pre-existing differences across markets.

Appendix Table A.6 reports the first-stage estimates at the bank-county-year level, based on the following specification:

$$\Delta\text{Deposits}_{b,c,t}/\text{Assets}_{b,t-1} = \beta Z_{b,c,t}/\text{Assets}_{b,t-1} + \theta_{b,t} + \lambda_{c,t} + \eta_{b,c} + \epsilon_{b,c,t} \tag{5}$$

where $\Delta\text{Deposits}_{b,c,t}$ represents the change in total deposits for bank b in county c from year $t-1$ to t, and $Z_{b,c,t}$ is the bank-county level deposit channel instrument defined in Equation (2). To mitigate the influence of outliers, both variables are scaled by lagged total bank assets. To account for unobserved heterogeneity, I include bank$\times$year fixed effects ($\theta_{b,t}$) and county$\times$year fixed effects ($\lambda_{c,t}$) in all specifications, with additional bank$\times$county fixed effects ($\eta_{b,c}$) included in the full sample. I cluster standard errors at both the bank and county levels to allow for multidimensional correlation in the error term. The model is estimated at an annual frequency, maintaining the same time alignment as the county-level analysis. The inclusion of county$\times$year fixed effects is important: it absorbs all variation common to a local market in a given year — including local credit demand, income shocks, and any market-wide flight-to-safety — so that identification comes from comparing banks with different deposit channel exposure within the same county and year. The instrument remains a strong predictor of deposit growth under this demanding specification, and notably does so for small banks (column 2), defined as those with total assets below $1 billion as of year-end 2008,

indicating that the deposits accrue across the local banking market rather than only to the largest institutions. As in the county-level results, the predictive power scales sharply with purchase intensity, rising from 0.31 in the QE I–III era (column 3) to 1.95 in the Pandemic QE era (column 4).

Table 2 presents the first-stage results at the bank level for both reserves and deposits. Because these variables are updated quarterly in the Call Reports, the model is estimated at the bank-quarter frequency, which yields cleaner, higher-frequency estimates than the annual SOD-based specifications above:

$$\Delta Y_{b,t} / \text{Assets}_{b,t-1} = \beta Z_{b,t} / \text{Assets}_{b,t-1} + \alpha_b + \gamma_t + \epsilon_{b,t} \tag{6}$$

where $\Delta Y_{b,t}$ in column (1) represents the quarterly change in bank-level reserves, while in columns (2) through (4) it denotes the quarterly change in bank-level deposits. To account for unobserved heterogeneity, I include bank fixed effects (α_b) and quarter fixed effects (γ_t). Standard errors are two-way clustered at the bank and state-quarter levels, where the latter is defined by the state in which the bank's headquarters is located. Columns (1) and (2) use an identical sample restricted to banks with non-missing reserve data, facilitating a direct comparison between the two liquidity measures. Column (3) reports results for the full sample of banks, and column (4) restricts this sample to small banks with total assets below $1 billion as of year-end 2008.

[Insert Table 2 Here]

At this quarterly frequency, the estimated pass-through is well below one: a $1 net security purchase allocated to a bank through the instrument is associated with roughly $0.30 of new deposits (columns 2 and 3), and $0.40 for small banks (column 4). Two features of the setting explain why the coefficient is below the theoretical benchmark of one. First, I measure purchases only through the eight domestic primary dealers and exclude foreign dealers, who also source securities from U.S. sellers; deposits created

through the foreign-dealer channel therefore appear in the dependent variable but not in the instrument, attenuating the coefficient. Second, the one-to-one benchmark holds only for purchases sourced from non-bank sellers; to the extent that the Fed purchases securities directly from banks' own holdings, the transaction is an asset swap that creates reserves but no new deposit, again pulling the coefficient below one. Column (1) shows that the deposit channel instrument also predicts reserve changes, with a coefficient of 0.21. This is notable because reserves are the exposure measure used in prior work such as Acharya et al. (2024): my instrument captures the same reserve variation they study, but it is not confined to it. Because a large share of banks — particularly smaller institutions — do not report reserves in their Call Reports, a reserve-based measure can be constructed only for the subset of banks that report them, whereas my deposit-based instrument applies to the full universe of deposit-taking institutions. By predicting both reserves and deposits, the instrument captures the variation that a reserve-based measure would pick up while extending measurement to the many banks for which reserves are unobserved. Reserves, in this sense, reflect only a partial view of the deposit channel — visible for the subset of banks that report them — whereas deposits capture the channel for the full universe of deposit-taking institutions. I focus the subsequent analysis on deposits for precisely this reason, as it allows me to capture the impact of the deposit channel across the broadest possible range of banking institutions.

Overall, the deposit channel instrument strongly predicts changes in deposits across all levels of aggregation — county, bank-county, and bank. The predictive power is concentrated in active QE periods and scales with the intensity of Federal Reserve purchases, indicating that QE-generated deposit inflows remain concentrated in the areas and institutions with high dealer exposure rather than being frictionlessly redistributed throughout the broader financial system. Collectively, these tests support the relevance condition.

3.3 Exclusion Restriction

The exclusion restriction cannot be formally tested, because the structural error terms affecting bank lending and local economic outcomes are unobservable. Nonetheless,

in this section I provide economic reasoning and auxiliary evidence to argue that the variation in the instrument is driven by the exogenous implementation of QE rather than by omitted variables or local shocks correlated with the supply of securities.

First, I address potential reverse causality within the shift-share framework. Because QE operations directly create deposits, using contemporaneous deposit shares to allocate liquidity would introduce a mechanical correlation between the instrument and the outcome. To make the shares pre-determined and plausibly purged of QE-induced effects, I fix all deposit shares at their mid-2008 levels.[11] Because QE had not yet commenced, these pre-determined weights reflect the structural geographic footprint of banks prior to the policy intervention.

Second, I rely on the aggregate exogeneity of the instrument. By construction, the instrument is anchored to the Federal Reserve's total net security purchases. These aggregate volumes reflect the Federal Reserve's macro-level policy mandates — specifically, targeting national inflation and employment — and are therefore unlikely to be influenced by the idiosyncratic credit demand of any specific bank or county.

Third, I address potential endogeneity in the cross-dealer allocation of purchases, which I term allocative exogeneity. A primary concern is that the volume of securities the Federal Reserve purchases from a specific primary dealer at a given time could be endogenous to the contemporaneous demand of that dealer's clients to sell those assets. For instance, in an area facing elevated local loan demand, the non-bank clients of a dealer operating there might sell more of their security holdings to fund local credit, raising that dealer's sales to the Fed at precisely the times when local lending is expanding. In such a scenario, the instrument could inadvertently capture demand-side factors correlated with the supply of securities, rather than a pure liquidity injection.

To test this possibility, I conduct a placebo test at the bank level. Specifically, I calculate each primary dealer's unconditional share of total Federal Reserve open market transactions over the 2009–2023 period, and construct an allocative-exogenous

[11]The key results are robust to using average deposit weights from 2005–2007 as an alternative pre-crisis benchmark.

instrument ($Z^{AE}_{b,t}$) using this time-invariant share in place of the dealer's period-specific purchase flows. Because this measure depends only on each dealer's long-run average magnitude, it cannot capture time-specific distributional shifts across dealers driven by fluctuating client demand. Column (1) of Table 3 shows that my baseline bank-level instrument ($Z_{b,t}$) is highly correlated with this allocative-exogenous instrument ($\rho = 0.93$). I then perform a Durbin-Wu-Hausman test, treating the baseline instrument as potentially endogenous and instrumenting it with the allocative-exogenous measure. As reported in Column (2) of Table 3, the test yields a p-value of 0.169, failing to reject the null hypothesis that the baseline instrument is exogenous. This indicates that the baseline measure is not significantly confounded by dealer-specific demand shocks, and that its variation derives from the stable, long-run distribution of purchases across dealers rather than from period-specific shifts that might reflect client demand.

[Insert Table 3 Here]

Finally, I control for persistent local factors. While one might worry that 2008 business shares correlate with long-term local demand trends, my empirical specifications include bank×year and county×year fixed effects. These fixed effects absorb time-varying heterogeneity in local credit demand and bank-specific supply shocks. Consequently, the remaining variation captured by the instrument reflects the interaction between the macro-level QE "shift" and the pre-determined "share," making it unlikely that the results are driven by factors other than exposure to QE-generated deposits.

4 The Deposit Channel of QE and Bank Lending

In this section, I study how the deposit channel of QE affects bank lending and compare it with the direct asset-swap channel, which mechanically exchanges securities for reserves at the headquarters level without generating new private deposits. I first employ bank-level regressions that incorporate my deposit channel instrument

alongside two Bartik-style instruments designed to capture the direct effects of the Treasury and MBS swaps. These additional instruments isolate the direct asset-swap component of QE's impact on bank lending. I then identify the deposit channel using bank-county-level regressions, which shut down the direct asset-swap effect and allow for a more precise estimate of the deposit channel. By exploiting this dual-level approach, I highlight the pivotal role of the deposit channel in driving credit supply and shaping real economic outcomes.

4.1 Direct Asset-Swap Channel vs. Deposit Channel

I begin by examining bank-level lending outcomes driven by both the direct asset-swap and the deposit channels of QE. Because the direct channel involves banks swapping eligible securities — specifically Treasuries and MBS — with the Federal Reserve, a bank's exposure to this channel is functionally constrained by its existing holdings of these assets. A bank's ex-ante position in these securities therefore determines its structural eligibility for the direct asset-swap mechanism.

To capture the direct effect, I construct two shift-share instruments by interacting a bank's ex-ante holdings of Treasuries and MBS with the aggregate Federal Reserve purchases of each asset class. These instruments track the reserve accumulation resulting from the direct exchange of securities for central bank liquidity, drawing on the logic of Rodnyansky and Darmouni (2017). A higher ex-ante concentration of these securities increases a bank's capacity to participate in the direct asset-swap channel. By incorporating these instruments, I can compare the magnitudes of the two channels of QE on bank lending. Crucially, while these instruments proxy for a bank's capacity to engage in the direct channel, they do not reflect its willingness to do so, so the coefficient on the direct asset-swap channel should be interpreted as the theoretical upper bound of the direct effect.

Specifically, I estimate the following bank-quarter-level regression:

$$\begin{aligned}\Delta Y_{b,t}/\text{Assets}_{b,t-1} = \beta_1 Z_{\text{tsy},b,t}/\text{Assets}_{b,t-1} + \beta_2 Z_{\text{mbs},b,t}/\text{Assets}_{b,t-1} \\ + \beta_3 Z_{b,t}/\text{Assets}_{b,t-1} + \alpha_b + \gamma_t + \epsilon_{b,t}\end{aligned} \tag{7}$$

where

$$Z_{\text{tsy},b,t} = \frac{\text{Treasury}_{b,t-1}}{\text{Treasury}_{t-1}} \times \text{Net Treasury Purchase}_t \tag{8}$$

and

$$Z_{\text{mbs},b,t} = \frac{\text{MBS}_{b,t-1}}{\text{MBS}_{t-1}} \times \text{Net MBS Purchase}_t \tag{9}$$

The results are reported in Table 4. Columns (1)–(3) report the effects of the direct asset-swap and deposit channels on total bank loans and on two primary subcategories, mortgage and commercial loans; column (4) reports the effect on securities holdings, which I use below to trace how banks fund the lending response. All specifications include bank (α_b) and quarter (γ_t) fixed effects, with robust standard errors two-way clustered at the bank and state-quarter levels as in Equation (6). The estimates suggest that bank lending is largely unaffected by the direct trading of Treasuries with the Federal Reserve. This lack of significance is consistent with the fact that commercial banks collectively hold only a small fraction of outstanding Treasury securities (see Appendix Figure A.2). In contrast, the direct effect stemming from MBS swaps and the deposit channel are comparable in magnitude: a \$1 increase in the direct QE reserve injection is associated with a \$0.24 increase in total lending, while a \$1 increase in the QE-driven deposit inflow is associated with a \$0.28 increase.

[Insert Table 4 Here]

A more granular look at lending types reveals a key distinction: the estimates indicate that the direct channel is associated with higher mortgage and C&I lending

contemporaneously, whereas the deposit channel is associated primarily with C&I lending, with no significant immediate effect on the mortgages banks retain on their balance sheets. This response is consistent with active portfolio reallocation under the direct channel: a bank that sells securities to the Fed may do so as part of a plan to redeploy the resulting reserves into higher-yielding assets such as long-term mortgage and commercial loans. Notably, my result on the direct MBS swap differs from Chakraborty, Goldstein and MacKinlay (2020), who find that MBS purchases crowd out commercial lending; I instead find that the active MBS swap is associated with a concurrent increase in both mortgage and commercial lending.[12]

The deposit channel's effect across loan types appears to reflect a difference in timing. A passive deposit inflow does not translate immediately into retained mortgages, for two reasons. Most mortgage originations are conforming loans that banks sell to the GSEs rather than hold, so a deposit inflow need not raise the stock of mortgages a bank retains on its balance sheet. The mortgages that banks do retain — principally jumbo loans — are long-duration assets, and a bank that receives a deposit inflow whose persistence it cannot yet judge has reason to build such a portfolio gradually rather than all at once. Commercial credit, by contrast, can be expanded more quickly on the margin, so the deposit channel shows up first in C&I lending. As deposits prove sticky, the bank becomes willing to lock the funding into longer-duration assets, and retained mortgage lending rises; I show in the following section that exogenous deposit inflows raise mortgage originations, and later that the effect is strongest for the balance-sheet-intensive jumbo segment.

I examine these dynamics directly using a local projection approach, tracing the impulse response of both the direct MBS-swap and deposit channels on future lending. Specifically, I estimate:

[12]The contrast with Chakraborty, Goldstein and MacKinlay (2020) likely reflects differences in sample period and specification rather than in the underlying mechanism. Their sample covers the QE I–III era, whereas mine extends through 2023 and incorporates the Pandemic QE episode, during which the Federal Reserve purchased large volumes of MBS under markedly different balance-sheet and credit-market conditions, so the direct MBS-swap effect on lending need not be stable across these regimes. In addition, my specification estimates the direct MBS-swap instrument jointly with the deposit channel instrument, so the MBS coefficient is identified holding deposit-driven lending fixed.

$$(Y_{b,t+h} - Y_{b,t-1})/\text{Assets}_{b,t-1} = \beta_{1h} Z_{\text{tsy},b,t}/\text{Assets}_{b,t-1} + \beta_{2h} Z_{\text{mbs},b,t}/\text{Assets}_{b,t-1} + \beta_{3h} Z_{b,t}/\text{Assets}_{b,t-1} + \alpha_b + \gamma_t + \epsilon_{b,t} \tag{10}$$

where Y represents mortgage and C&I lending in Panels A and B of Appendix Figure A.3, respectively, and $h = 0, 1, \ldots, 7$ indexes the projection horizon in quarters. The projections are suggestive of the timing interpretation. The deposit channel's effect on retained mortgages appears essentially zero on impact and turns positive only after roughly three quarters, building gradually thereafter. Its effect on commercial lending, by contrast, rises more quickly, appears to peak around the third quarter, and then recedes. One reading of this pattern is that commercial credit serves as the initial home for funding whose permanence is still uncertain, and that as that uncertainty resolves the bank may rotate the funding out of short-maturity commercial credit and into longer-duration mortgages, so that the decline in commercial lending and the build-up in retained mortgages could be, at least in part, two sides of the same reallocation. Viewed this way, the deposit channel may offer a complementary account of the mortgage-for-commercial substitution that Chakraborty, Goldstein and MacKinlay (2020) document: where they identify banks crowding out C&I lending as they expand mortgage originations, the dynamics here are consistent with a similar reallocation operating through the liability side, as banks gradually shift QE-generated funding from commercial credit toward retained mortgages.

This reallocation is also visible on the rest of the balance sheet. Column (4) of Table 4 shows that a deposit inflow is associated with a decline in securities holdings. Combining this with the first-stage estimates in Table 2, the deposit channel's balance-sheet response approximately accounts for itself. A \$1 net purchase allocated to a bank through the instrument is associated with roughly \$0.32 of new deposits (Table 2, column 3) on the liability side. On the asset side, the same \$1 is associated with roughly \$0.21 of additional reserves (Table 2, column 1) and \$0.28 of additional loans (Table 4, column 1), partly offset by a \$0.18 decline in securities (Table 4, column

4). The asset-side responses therefore sum to approximately the \$0.32 deposit inflow ($0.21 + 0.28 - 0.18 = 0.31$), so the balance sheet expands by roughly the size of the funding shock. This decomposition suggests that banks fund the lending and reserve expansion not only from the new deposits but also by drawing down their securities portfolios. Such behavior is consistent with securities serving as a liquidity buffer that banks deploy when a funding inflow coincides with attractive lending opportunities; to the extent that the accompanying reserves consume balance-sheet capacity under the leverage ratio (Diamond, Jiang and Ma, 2024), it is also consistent with banks reallocating that capacity away from securities and toward higher-yielding loans.

Overall, these findings indicate that the asset-reallocation responses to active direct asset swaps and to passive deposit inflows are both significant in magnitude but differ in loan category and in timing. To my knowledge, this provides the first empirical evidence quantifying the relative magnitudes of these two transmission mechanisms in driving bank credit supply.

4.2 Isolating the Deposit Channel

To isolate the deposit channel of QE from the direct asset-swap channel, I turn to a bank-county-level analysis. I examine two types of lending that banks are required to report at the county-year level: mortgage loans under the Home Mortgage Disclosure Act (HMDA) and small business loans under the Community Reinvestment Act (CRA). This granular approach allows for the inclusion of bank$\times$year fixed effects ($\theta_{b,t}$), which compare lending across different counties served by the same bank in the same year.

Because a bank's reserve account is centrally managed by its treasury department, variation in lending across the different counties served by the same bank cannot be attributed to the bank's aggregate reserve position. The direct channel of QE — the asset swap resulting in a centralized reserve increase — is therefore absorbed by the bank$\times$year fixed effects. QE-generated deposits, by contrast, arrive disproportionately at different locations within the same bank's network and tend to remain local. This implies that the remaining variation is plausibly driven by the deposit channel rather than

the centralized reserve position, with local credit demand absorbed by the county×year fixed effects. I estimate the following specification:

$$\Delta Y_{b,c,t} / \text{Assets}_{b,t-1} = \beta_1 Z_{b,c,t} / \text{Assets}_{b,t-1} + \theta_{b,t} + \lambda_{c,t} + \eta_{b,c} + \epsilon_{b,c,t} \tag{11}$$

where the bank×year fixed effects ($\theta_{b,t}$) absorb the centralized asset-swap channel and any time-varying bank-level credit supply, the county×year fixed effects ($\lambda_{c,t}$) absorb local credit demand, and the bank×county fixed effects ($\eta_{b,c}$) absorb persistent, bank-specific lending specializations across counties.

The unit of observation is a bank-county-year. I begin with all bank-county-years in which the bank reports activity in the HMDA or CRA data. I restrict the sample to counties within Metropolitan Statistical Areas (MSAs), where the instrument is not mechanically zero.[13] My primary sample further restricts attention to "in-market" bank-counties — those in which the bank maintains a physical deposit-taking presence in the county — so that the instrument reflects genuine variation in QE-generated deposit inflows across the markets in which the bank actually operates and takes deposits. I drop bank-counties in which the relevant lending variable is missing; in particular, a bank-county enters the mortgage regression only if the bank originates mortgages in that county, and enters the small business regression only if the bank reports small business lending there. To compare the two loan types on a common footing, columns (1) and (2) of Table 5 are estimated on the common set of bank-counties in which the bank originates both mortgage and small business loans, with column (1) reporting mortgage lending and column (2) small business lending on this shared sample. Column (3) instead uses all in-market bank-counties with small business lending, whether or not the bank also originates mortgages there.

Table 5 reports the results. For a \$1 QE-generated deposit inflow, banks increase mortgage lending by \$0.25 (column 1), compared with a \$0.09 increase in small business

[13] Primary dealers have very limited coverage outside MSAs, so the deposit channel instrument is mechanically zero or near-zero for most non-MSA counties. Restricting the sample to MSAs therefore drops observations that carry no identifying variation rather than discarding informative data.

lending on the same set of bank-counties (column 2). Banks therefore expand mortgage lending relative to small business lending by a factor of roughly three to one in response to the deposit channel. One interpretation is that mortgages are highly standardized, whereas small business loans typically require soft information and specialized risk assessment, so banks face higher frictions when expanding small-business portfolios rapidly following a sudden deposit inflow.

[Insert Table 5 Here]

In Appendix Table A.7, I report the broader sample that also includes "out-of-market" bank-counties — those in which the bank originates loans but has no deposit-taking presence, and for which the instrument is therefore zero by construction. I include an indicator, In Branch, for whether the bank has a physical presence in the county. The deposit channel coefficients are larger in this broader sample (for example, $1.57 for mortgage lending), but I treat the in-market sample as my preferred specification: it compares counties in which the bank genuinely takes deposits, so the identifying variation reflects differences in local deposit inflows rather than the mechanical contrast between deposit-taking and non-deposit-taking counties. The two samples therefore differ in magnitude, but both point to a substantially larger response of mortgage than small business lending to the deposit channel.

5 The Deposit Channel of QE and Employment Growth

In the previous section, I showed that banks increase C&I lending in areas where they receive large QE-generated deposit inflows. Because commercial lending generally supports greater business activity and hiring, I test in this section whether the deposit channel of QE exerts real effects on local employment growth. By incorporating county-level exposure measures for both the direct asset-swap and deposit channels, I isolate and compare their respective effects on local employment. To account for the limited employment in rural markets, I restrict the sample to counties within MSAs.

To capture the direct asset-swap effect of QE, I construct two county-level instruments that measure the reserve expansion resulting from banks' capacity to sell Treasury and MBS holdings to the Federal Reserve. These are calculated as the deposit-weighted average of the bank-level instruments defined in Equations (8) and (9) across all banks operating within a given county. To ensure these measures are not confounded by contemporaneous QE-driven deposit shifts, I fix the weights using each bank's pre-QE (2008) deposit share within the county. These aggregated instruments proxy for the volume of QE-eligible securities held by the local banking system, thereby capturing the geographic intensity of the direct asset-swap channel.

$$Z_{\text{tsy},c,t} = \sum_{b \in c} \frac{\text{Deposits}_{b,c}}{\text{Deposits}_{c}} \times Z_{\text{tsy},b,t} \tag{12}$$

$$Z_{\text{mbs},c,t} = \sum_{b \in c} \frac{\text{Deposits}_{b,c}}{\text{Deposits}_{c}} \times Z_{\text{mbs},b,t} \tag{13}$$

I estimate a distributed-lag model for quarterly county-level employment growth that includes the contemporaneous instrument and its seven quarterly lags, estimated separately for each QE episode:

$$\begin{aligned}\Delta \ln(Y)_{c,t} = &\sum_{j=0}^{7} \beta_{1j} \text{STD}(Z_{\text{tsy},c,t-j}/\text{Deposits}_{c,t_0}) + \sum_{j=0}^{7} \beta_{2j} \text{STD}(Z_{\text{mbs},c,t-j}/\text{Deposits}_{c,t_0}) \\ &+ \sum_{j=0}^{7} \beta_{3j} \text{STD}(Z_{c,t-j}/\text{Deposits}_{c,t_0}) + \delta_c + \gamma_t + \epsilon_{c,t}\end{aligned} \tag{14}$$

where Y represents total, tradable-sector, and non-tradable-sector county employment in columns (1), (2), and (3) of Table 6, respectively, and $\Delta \ln(Y)_{c,t} = \ln(Y_{c,t}) - \ln(Y_{c,t-1})$ is the quarterly log change in employment. I adopt the industry classification of Mian and Sufi (2014) to categorize industries into tradable and non-tradable sectors. The instruments are scaled by county-level deposits ($t_0 = 2008$ for the QE I–III era and $t_0 = 2019$ for the Pandemic QE era) and standardized, so that each coefficient is

interpreted as the effect of a one-standard-deviation increase in exposure on quarterly employment growth. I estimate the model separately for the QE I–III era (Panel A) and the Pandemic QE era (Panel B). The contemporaneous-plus-seven-quarter lag structure allows the employment response to build over roughly two years, consistent with the time required for business expansion and hiring. The model includes county fixed effects (δ_c) and quarter fixed effects (γ_t), and standard errors are two-way clustered at the county and state-quarter levels. The coefficients reported in Table 6 are cumulative dynamic multipliers — the sum of the contemporaneous and the seven lagged terms — reflecting the total effect of QE exposure on employment growth.

[Insert Table 6 Here]

Table 6 shows that the deposit channel of QE is associated with higher local employment growth, concentrated in the non-tradable sector. Because the regressors are standardized and the dependent variable is a log change in employment, each coefficient measures the cumulative change in employment associated with a one-standard-deviation increase in exposure to the channel. For non-tradable employment (column 3), a one-standard-deviation increase in deposit channel exposure is associated with cumulative employment growth higher by 0.005 during the QE I–III era (Panel A) and by 0.027 during the Pandemic QE era (Panel B), both significant at the one percent level. By contrast, the deposit channel shows no economically meaningful relationship with tradable employment (column 2), where the estimates are small and statistically insignificant in both eras. This sectoral pattern is what a local demand mechanism would predict: following Mian and Sufi (2014), non-tradable employment responds to local demand shocks because non-tradable goods and services must be produced and consumed locally, whereas tradable employment is tied to national or global demand and is therefore less sensitive to a localized liquidity injection.

The effect of the deposit channel on total employment (column 1) is small in both eras ($\beta = 0.000$ in the QE I–III era and $\beta = 0.003$ in the Pandemic QE era, the latter

significant). This is consistent with the sectoral decomposition: total employment pools the non-tradable sector together with tradable and other industries, so the non-tradable response is diluted in the aggregate. The decomposition in columns (2) and (3) is therefore more informative about the channel than the total, and it locates the effect in the non-tradable sector, where a local demand mechanism would operate.

The direct asset-swap channel shows a less consistent relationship with employment across eras and sectors. In the QE I–III era, the direct Treasury-swap instrument is associated with higher tradable and non-tradable employment, while the direct MBS-swap instrument enters with mixed signs and is largely insignificant; in the Pandemic QE era, neither direct instrument is reliably related to employment. The deposit channel, by contrast, is consistently associated with higher non-tradable employment in both eras. Taken together, these results are consistent with the deposit channel of QE stimulating the local economy by raising regional demand, generating an increase in employment concentrated in non-tradable industries.

6 The Deposit Channel of QE and Local House Prices

In the previous sections, I showed that banks significantly increase C&I lending — fostering greater local business activity and higher employment growth in the non-tradable sector — as well as mortgage lending, in areas that receive large QE-generated deposit inflows. These shifts in both credit supply and labor demand are likely to stimulate local housing markets. To test whether the deposit channel of QE exerts real effects on the housing sector, I again use the deposit channel instrument at the county level. By incorporating county-level exposure measures for both the direct asset-swap and deposit channels, I isolate and compare their respective impacts on local house prices and total mortgage activity.

I begin by examining price effects using two standard metrics: the FHFA House Price Index (HPI) and the Zillow Home Value Index (ZHVI). Recognizing that these indices often rely on appraisal-based valuations and model-driven estimates, I further leverage granular HMDA data to examine the impact of the deposit channel on realized

home-purchase transactions. To account for the infrequent sales and limited lending activity characteristic of rural markets, I restrict the sample to counties within MSAs, covering approximately 1,700 counties across the United States.

6.1 Effect on House Prices

I first estimate the impact of the deposit channel on local house prices using indices provided by the FHFA (HPI) and Zillow (ZHVI). I estimate the following specification for the annual FHFA HPI, reported in columns (1) and (2) of Table 7:

$$\begin{aligned}\Delta \ln(\text{HPI})_{c,t} = &\sum_{j=0}^{1} \beta_{1j} \text{STD}(Z_{\text{tsy},c,t-j}/\text{Deposits}_{c,t_0}) + \sum_{j=0}^{1} \beta_{2j} \text{STD}(Z_{\text{mbs},c,t-j}/\text{Deposits}_{c,t_0}) \\ &+ \sum_{j=0}^{1} \beta_{3j} \text{STD}(Z_{c,t-j}/\text{Deposits}_{c,t_0}) + \delta_c + \gamma_t + \epsilon_{c,t}\end{aligned} \tag{15}$$

and for the quarterly Zillow ZHVI, reported in columns (3) and (4):

$$\begin{aligned}\Delta \ln(\text{ZHVI})_{c,t} = &\sum_{j=0}^{4} \beta_{1j} \text{STD}(Z_{\text{tsy},c,t-j}/\text{Deposits}_{c,t_0}) + \sum_{j=0}^{4} \beta_{2j} \text{STD}(Z_{\text{mbs},c,t-j}/\text{Deposits}_{c,t_0}) \\ &+ \sum_{j=0}^{4} \beta_{3j} \text{STD}(Z_{c,t-j}/\text{Deposits}_{c,t_0}) + \delta_c + \gamma_t + \epsilon_{c,t}\end{aligned} \tag{16}$$

The FHFA regressions are estimated at the county-year level and include county and year fixed effects, with standard errors two-way clustered at the county and state-year levels. The Zillow regressions are estimated at the county-quarter level and include county and quarter fixed effects, with standard errors two-way clustered at the county and state-quarter levels. In both specifications, all independent variables are scaled by 2008 county-level deposits for the QE I–III sample and by 2019 deposits for the Pandemic QE sample, and are then standardized so that each coefficient is interpreted as the percentage change in house prices associated with a one-standard-deviation increase in exposure. Recognizing that house prices may respond to QE-injected liquidity with a

lag, I include both the contemporaneous instrument and its one-year lag — equivalent to four quarterly lags in the ZHVI specifications — to capture delayed effects, and the coefficients reported in Table 7 represent the sum of the contemporaneous and lagged terms. Because the FHFA specifications are annual and the Zillow specifications are quarterly, the two sets of coefficients are not directly comparable in levels: the quarterly estimates measure per-quarter growth and must be cumulated over four quarters to be placed on an annual basis.

[Insert Table 7 Here]

The results show a consistent and economically large effect of the deposit channel on local house prices. In the annual FHFA specifications, a one-standard-deviation increase in deposit channel exposure raises house-price growth by 1.9 percentage points during the QE I–III era (column 1, $\beta = 0.019$) and 1.1 percentage points during the Pandemic QE era (column 2, $\beta = 0.011$). In the quarterly Zillow specifications, the corresponding coefficients are 0.004 (column 3) and 0.006 (column 4) per quarter; cumulated over four quarters, these imply annual effects of roughly 1.6 and 2.4 percentage points, respectively. The two independent data sources therefore deliver mutually consistent magnitudes: across both indices and both episodes, a one-standard-deviation increase in deposit channel exposure is associated with an annual house-price increase on the order of 1 to 2.5 percentage points.

A notable feature of the results is the stability of this effect across the two QE eras. It is widely recognized that the Pandemic QE era coincided with rapid house-price inflation, and the deposit channel is clearly operative then — a one-standard-deviation increase in exposure raises annual house-price growth by 1.1 percentage points in the FHFA data and roughly 2.4 percentage points in the Zillow data. Less appreciated is that the same channel was already at work during the QE I–III era, when the FHFA effect was if anything larger (1.9 percentage points). The deposit channel therefore does not merely describe the pandemic-era housing boom; it identifies a transmission mechanism

that has operated consistently across QE programs separated by more than a decade and conducted under very different macroeconomic conditions. This consistency is difficult to attribute to era-specific factors and instead points to a structural feature of how QE reaches local housing markets.

By comparison, the direct asset-swap channel has a positive but smaller and less consistent effect on house prices. The direct MBS-swap coefficient is statistically significant in all four specifications, though it is negative in one (the QE I–III Zillow column), and the deposit channel coefficient exceeds it in three of the four; the one exception is the Pandemic-era FHFA column, where the direct MBS effect is larger. The direct Treasury-swap instrument shows no significant relationship in any specification, consistent with banks' negligible Treasury holdings documented earlier. On balance, this pattern suggests that QE transmits to local house prices substantially through the redistribution of liquidity via deposit inflows, alongside the direct exchange of securities from local bank portfolios for reserves.

6.2 Effect on Mortgage Lending Activity

To address potential concerns regarding the use of house price indices — specifically the reliance on appraisal-based valuations, model-driven estimates, and the selective coverage of conforming mortgages — I supplement the analysis with comprehensive loan-level data from HMDA. By shifting the focus to realized mortgage lending activity, I can more directly observe the flow of credit that accompanies house price appreciation in areas with varying exposure to the deposit channel of QE.

To capture broad trends in the residential housing market and to maintain consistency with the existing literature, I apply several standard filters to the HMDA data.[14] Specifically, I restrict the sample to originated, first-lien, conventional mortgages for owner-occupied, one-to-four-family properties, and I exclude Home Ownership and Equity Protection Act (HOEPA) loans, reverse mortgages, and loans intended for commercial purposes. Because the goal is to isolate the credit supply that accompanies house price appreciation, I further restrict the sample to home-purchase originations

[14] See, e.g., Amornsiripanitch et al. (2024).

and exclude refinancings, whose volume is driven primarily by movements in the national mortgage rate rather than by local credit supply. These restrictions facilitate a cleaner identification of credit supply effects within the primary residential market.

I aggregate home-purchase originations to the county-year level and focus on two measures of local credit supply: the annual number of loans originated and their aggregate dollar volume. Because HMDA records the flow of new originations each year rather than a stock of loans outstanding, the dependent variable is the (log) annual origination flow; the county fixed effects absorb each county's baseline level of originations, so the coefficients capture proportional differences in new lending across counties with different exposure. To allow the lending response to build over time, I estimate a distributed-lag specification that includes the contemporaneous instruments and their one-year lags:

$$\ln(Y)_{c,t} = \sum_{j=0}^{1} \beta_{1j} \text{STD}(Z_{\text{tsy},c,t-j}/\text{Deposits}_{c,t_0}) + \sum_{j=0}^{1} \beta_{2j} \text{STD}(Z_{\text{mbs},c,t-j}/\text{Deposits}_{c,t_0}) + \sum_{j=0}^{1} \beta_{3j} \text{STD}(Z_{c,t-j}/\text{Deposits}_{c,t_0}) + \delta_c + \gamma_t + \epsilon_{c,t} \tag{17}$$

where $Y_{c,t}$ is the annual number of home-purchase originations (columns 1 and 3 of Table 8) or their aggregate dollar volume (columns 2 and 4), entered in logs. The instruments are scaled by county-level deposits ($t_0 = 2008$ for the QE I–III era and $t_0 = 2019$ for the Pandemic QE era) and standardized, so that each coefficient is the approximate percentage difference in annual originations associated with a one-standard-deviation increase in exposure. I include the Treasury and MBS instruments alongside the deposit channel instrument to absorb the direct asset-swap effect, so that the estimated deposit channel effects are not confounded by the simultaneous asset-swap channel. All specifications include county and year fixed effects, with standard errors two-way clustered at the county and state-year levels. The reported coefficients are cumulative dynamic multipliers — the sum of the contemporaneous and one-year

lagged terms. Columns (1)–(2) report the QE I–III era and columns (3)–(4) the Pandemic QE era.

[Insert Table 8 Here]

Table 8 shows that the deposit channel raises local home-purchase mortgage activity in both QE episodes. A one-standard-deviation increase in deposit channel exposure is associated with roughly 2.8% more purchase originations by count ($\beta = 0.028$) and 2.9% more by dollar volume ($\beta = 0.029$) during the QE I–III era, and with roughly 4.7% more by count ($\beta = 0.047$) and 4.3% more by volume ($\beta = 0.043$) during the Pandemic QE era, all significant at the one percent level. The effect is therefore present and economically meaningful in both periods, and somewhat larger during the pandemic. This consistency mirrors the house-price results in Table 7: the deposit channel raises both house prices and local purchase lending across two QE programs conducted under very different macroeconomic conditions.

The effects on purchase lending are also larger than the house-price effects in Table 7 — for example, the 4.3% Pandemic-era volume effect exceeds the roughly 2% house-price effect — a difference consistent with the broader market coverage of the HMDA data. The FHFA HPI is restricted to homes financed by conforming mortgages, and the ZHVI targets the 35th–65th percentile of the price distribution, whereas the HMDA-based measures capture originations across the full market, including the jumbo and higher-priced segments where the deposit channel operates most strongly. The next section examines this jumbo segment directly.

The direct asset-swap channel plays a more limited role in local purchase lending. The direct MBS-swap instrument is positive and significant in the QE I–III era ($\beta = 0.015$ by count and 0.023 by volume) but is not statistically significant in the Pandemic QE era, while the direct Treasury-swap instrument shows no significant relationship in either period, consistent with banks' negligible Treasury holdings documented earlier. Across specifications, the deposit channel is the more consistently significant of the two, which

indicates that QE transmits to local purchase-mortgage activity primarily through the geographic incidence of deposit creation rather than through the direct exchange of securities from local bank portfolios for reserves.

7 The Deposit Channel and Balance-Sheet Expansion

The deposit channel of QE is characterized by an expansion of banks' balance sheets, which enlarges their capacity to lend and therefore produces a credit supply effect with wide implications. This credit supply effect is already visible in Table 5, where the county×year fixed effects absorb local credit demand and identify a supply-driven increase in lending.

In this section, I draw out two further implications of this credit supply effect. First, I show that banks more exposed to the deposit channel raise their mortgage approval rates and retain a larger share of mortgages on their balance sheets — a direct manifestation of asset-side balance-sheet expansion. Second, I find that the effect is more pronounced for jumbo mortgages, which are more difficult to securitize through the GSEs and are therefore more sensitive to bank balance-sheet capacity than conforming loans. Because this balance-sheet expansion is a distinctive byproduct of the deposit channel, these findings underscore the importance of accounting for the credit supply consequences of QE.

7.1 Mortgage Approval and Retention

I estimate the following specification at the bank-county-year level, separately for each QE episode:

$$Y_{b,c,t} = \beta_1 \text{STD}(Z_{b,c,t} / \text{Assets}_{b,t_0}) + \beta_2 \text{In Branch}_{b,c,t} + \theta_{b,t} + \lambda_{c,t} + \eta_{b,c} + \epsilon_{b,c,t} \qquad (18)$$

where $Y_{b,c,t}$ is the mortgage approval rate in column (1), the log of the annual number of retained mortgages in column (2), and the log of their annual dollar volume in column (3) of Table 9; as with the origination measures, the retained-mortgage variables are

annual flows, and the bank×county fixed effects absorb each bank-county's baseline level of retention. The instrument is scaled by pre-period bank assets ($t_0 = 2008$ for the QE I–III era and $t_0 = 2019$ for the Pandemic QE era). The county×year fixed effects ($\lambda_{c,t}$) absorb local credit demand, isolating a supply-side effect; the bank×county fixed effects ($\eta_{b,c}$) and the physical-branch indicator control for persistent local lending patterns. The analysis is restricted to MSAs, where the deposit channel instrument is more precisely measured. Panel A of Table 9 reports the QE I–III era and Panel B the Pandemic QE era.

[Insert Table 9 Here]

Banks can expand mortgage credit on the extensive margin by approving a larger share of applications. Column (1) of Table 9 shows that they do, in both eras: a one-standard-deviation increase in deposit channel exposure raises the mortgage approval rate by 0.1 percentage points in the QE I–III era ($\beta = 0.001$) and by 0.3 percentage points in the Pandemic QE era ($\beta = 0.003$), both statistically significant. Because the unconditional approval rate in both samples is roughly 90% — equivalently, a rejection rate near 10% — these increases represent about a 1% reduction in the rejection rate in the QE I–III era and about a 3% reduction during the Pandemic QE era. The effect operates at the margin where credit constraints bind, which suggests that the deposit channel eases access to mortgage credit for marginal borrowers rather than merely expanding lending to those who would have been approved regardless.

The deposit channel should also shift banks toward retaining mortgages, not merely originating them for sale. Although banks routinely originate conforming mortgages to securitize through the GSEs, an inflow of QE-generated deposits creates an incentive to expand the asset side of the balance sheet by holding more loans on book. Columns (2) and (3) of Table 9 are consistent with this in both eras. A one-standard-deviation increase in deposit channel exposure is associated with approximately 0.9% more retained mortgages by count ($\beta = 0.009$) and 0.8% more by dollar volume ($\beta = 0.008$) in the QE I–III era, and with approximately 3% more by count ($\beta = 0.030$) and 3.5% more

by volume ($\beta = 0.035$) in the Pandemic QE era. The effect is present in both periods and larger during the pandemic, indicating a shift in bank portfolios toward retained real estate assets.

7.2 Conforming vs. Jumbo Loans

I now examine how the deposit channel's effect differs across mortgage segments with differing balance-sheet sensitivity: conforming versus jumbo loans. Conforming loans have principal balances below the FHFA Conforming Loan Limit (CLL) and are eligible for purchase by the GSEs; because they can be readily securitized and moved off balance sheet, their origination is less constrained by the originating bank's capital position. Jumbo loans exceed the limit, are ineligible for GSE purchase, and are therefore typically retained on balance sheet, which makes the supply of jumbo credit highly sensitive to internal balance-sheet capacity. Because the deposit channel operates precisely through balance-sheet expansion, it should have a stronger effect on jumbo origination than on conforming origination. I restrict this analysis to the Pandemic QE period, because the conforming-versus-jumbo classification requires the loan-level conforming-loan-limit flag, which became available only with the 2018 expansion of HMDA reporting.

I estimate the following specification:

$$\ln(Y)_{b,c,t} = \beta_1 \text{STD}(Z_{b,c,t}/\text{Assets}_{b,2019}) + \beta_2 \text{In Branch}_{b,c,t} + \theta_{b,t} + \lambda_{c,t} + \eta_{b,c} + \epsilon_{b,c,t} \quad (19)$$

where $Y_{b,c,t}$ is the annual flow of total, conforming, and jumbo mortgage originations in columns (1), (2), and (3) of Table 10, respectively, entered in logs; Panel A measures originations by count and Panel B by dollar volume. As with the earlier mortgage measures, the dependent variable is the log of an annual origination flow, and the bank×county fixed effects absorb each bank-county's baseline level of originations. I control for bank×county fixed effects ($\eta_{b,c}$) and a physical-branch indicator to absorb persistent local lending patterns, while the bank×year fixed effects ($\theta_{b,t}$) absorb the

direct asset-swap effect, so that the coefficients identify the deposit channel across mortgage segments.

[Insert Table 10 Here]

Table 10 reports the results, with Panels A and B measuring the number and the dollar volume of originations, respectively. The pattern matches the balance-sheet mechanism: a one-standard-deviation increase in deposit channel exposure is associated with more jumbo origination ($\beta = 0.033$ in both panels) than conforming origination ($\beta = 0.024$ by count and 0.028 by volume), even though conforming lending is far larger in absolute terms. That the more balance-sheet-intensive segment responds more strongly is consistent with a credit supply effect operating through bank funding capacity rather than through borrower demand, which would not single out the retained segment. Taken together with the approval-rate and retention results, these findings provide evidence of a credit supply effect driven by the deposit channel of QE.

8 Conclusion

This paper studies the transmission of quantitative easing through the liability side of bank balance sheets. When the Federal Reserve purchases securities from non-bank sellers, it creates private-sector deposits that accrue disproportionately to banks in the financial centers where institutional securities trading is concentrated. Using a shift-share instrument that interacts the Fed's net purchase flows with the pre-QE geographic footprint of primary dealers, I find that this deposit channel of QE is quantitatively comparable to the direct asset-swap channel, reaches a far broader universe of banks, and — because the deposits it creates are geographically concentrated — generates heterogeneous local effects on credit, employment, and house prices. The channel raises bank lending and tilts credit toward mortgages, particularly the jumbo loans banks must retain on balance sheet, and stimulates local non-tradable employment and house prices. Its house-price effect is notably stable across the QE I–III and pandemic-era

programs, suggesting a structural feature of how QE reaches housing markets rather than an artifact of any single episode.

These results carry implications for how QE is understood and evaluated. Because the deposit channel operates independently of a bank's securities holdings, analyses that measure QE exposure only through asset or reserve positions may understate both its reach and its real consequences, particularly for the many smaller banks that hold few eligible securities. And because the channel concentrates liquidity and credit in particular local markets, QE carries distributional consequences across regions that aggregate analyses do not capture — a footprint relevant not only for evaluating past programs but for anticipating that of future ones.

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Table 1: Primary Dealers of the Federal Reserve Bank of New York

This table lists the 26 primary dealers as of February 2026. These institutions serve as the New York Fed's counterparties for open market operations and Treasury auctions.

BofA Securities, Inc.	Bank of Montreal, Chicago Branch
Citigroup Global Markets Inc.	Bank of Nova Scotia, New York Agency
J.P. Morgan Securities LLC	Barclays Capital Inc.
Morgan Stanley & Co. LLC	BNP Paribas Securities Corp.
Wells Fargo Securities, LLC	Daiwa Capital Markets America Inc.
Jefferies LLC	Deutsche Bank Securities Inc.
Cantor Fitzgerald & Co.	HSBC Securities (USA) Inc.
ASL Capital Markets Inc.	Mizuho Securities USA LLC
Goldman Sachs & Co. LLC	MUFG Securities Americas Inc.
Nomura Securities International	NatWest Markets Securities Inc.
RBC Capital Markets, LLC	SMBC Nikko Securities America, Inc.
Santander US Capital Markets LLC	Societe Generale, New York Branch
TD Securities (USA) LLC	UBS Securities LLC

Table 2: First-Stage Instrument Relevance for Bank Reserves and Deposits

This table reports first-stage regression results at the bank-quarter level. The dependent variables are the quarterly change in bank-level reserves in column (1) and the quarterly change in bank-level deposits in columns (2) through (4). Columns (1) and (2) share the same sample of banks that report reserves. Column (3) reports deposit changes for all banks with available deposit data. Column (4) reports deposit changes for small banks, defined as those with total assets below $1 billion as of year-end 2008. The sample period covers 2009–2023. Both the dependent variables and the deposit channel instrument are scaled by the previous quarter's total bank-level assets and winsorized at the 1st and 99th percentiles. All specifications include bank and quarter fixed effects. Robust standard errors, two-way clustered at the bank and state-quarter levels, are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta\text{Reserves}_{b,t}/\text{Assets}_{b,t-1}$	$\Delta\text{Deposits}_{b,t}/\text{Assets}_{b,t-1}$		
$Z_{b,t}/\text{Assets}_{b,t-1}$	0.21***	0.30***	0.32***	0.40***
	(0.04)	(0.08)	(0.06)	(0.07)
N	98,617	98,617	356,265	307,868
R^2	0.06	0.16	0.15	0.14
F-stat	24.36	14.79	24.55	31.38
Bank FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes
Bank Size	All	All	All	Small
Sample Period	Full Sample	Full Sample	Full Sample	Full Sample

Table 3: Allocative Exogeneity of the Deposit Channel Instrument

This table reports IV regression results evaluating the allocative exogeneity of the bank-level deposit channel instrument. The allocative-exogeneity instrument, $Z^{\text{AE}}_{b,t}$, is constructed using each primary dealer's time-invariant, unconditional share of total Federal Reserve open market transactions. Both the baseline deposit channel instrument ($Z_{b,t}$) and the allocative-exogeneity instrument ($Z^{\text{AE}}_{b,t}$) are scaled by lagged total bank assets. Column (1) reports the first-stage regression, where $Z_{b,t}$ is regressed on $Z^{\text{AE}}_{b,t}$ to establish relevance. Column (2) reports the second-stage results of a Durbin-Wu-Hausman test, in which $Z_{b,t}$ is treated as potentially endogenous and instrumented by $Z^{\text{AE}}_{b,t}$. The sample spans 2009–2023, with all variables winsorized at the 1st and 99th percentiles. All specifications include bank and quarter fixed effects. Robust standard errors, two-way clustered at the bank and state-quarter levels, are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)
Dependent Variable:	$Z_{b,t}/\text{Assets}_{b,t-1}$	$\Delta\text{Deposits}_{b,t}/\text{Assets}_{b,t-1}$
$Z^{\text{AE}}_{b,t}/\text{Assets}_{b,t-1}$	0.93***	
	(0.01)	
$Z_{b,t}/\text{Assets}_{b,t-1}$		0.28***
		(0.07)
N	356,265	356,265
F-stat	6,365.97	13.93
Durbin-Wu-Hausman p-Value		0.169
Bank FE	Yes	Yes
Quarter FE	Yes	Yes
IV Stage	First	Second
Sample Period	Full Sample	Full Sample

Table 4: Direct Asset-Swap vs. Deposit Channel: Effects on Bank Assets

This table reports regression results examining the effect of the direct asset-swap and deposit channels of QE on bank-level asset outcomes. The dependent variables in columns (1)–(3) are the quarterly changes in total loans, mortgage loans, and commercial and industrial (C&I) loans, respectively, and in column (4) the quarterly change in securities holdings; each is normalized by lagged total assets. The primary regressors are a bank-level deposit channel instrument and two shift-share instruments reflecting the direct asset-swap effect; all are normalized by the previous quarter's total bank-level assets for comparability. The sample covers 2009–2023, with all variables winsorized at the 1st and 99th percentiles. All specifications include bank and quarter fixed effects. Robust standard errors, two-way clustered at the bank and state-quarter levels, are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Dependent Variable:	$\frac{\Delta \text{Loans}_{b,t}}{\text{Assets}_{b,t-1}}$	$\frac{\Delta \text{Mortgages}_{b,t}}{\text{Assets}_{b,t-1}}$	$\frac{\Delta \text{C\&I}_{b,t}}{\text{Assets}_{b,t-1}}$	$\frac{\Delta \text{Securities}_{b,t}}{\text{Assets}_{b,t-1}}$
$Z_{\text{tsy},b,t} / \text{Assets}_{b,t-1}$	2.53	0.08	-0.30	-0.97
	(2.52)	(0.83)	(0.91)	(1.84)
$Z_{\text{mbs},b,t} / \text{Assets}_{b,t-1}$	0.24***	0.05***	0.09***	-0.18***
	(0.05)	(0.02)	(0.02)	(0.05)
$Z_{b,t} / \text{Assets}_{b,t-1}$	0.28***	-0.00	0.10***	-0.18***
	(0.05)	(0.01)	(0.02)	(0.02)
N	356,264	356,264	356,264	356,264
R^2	0.22	0.14	0.20	0.08
Bank FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes
Sample Period	Full Sample	Full Sample	Full Sample	Full Sample

Table 5: The Deposit Channel of QE: Effects on In-Market Bank-County Lending

This table reports regression results examining the effect of the deposit channel of QE on bank-county lending within Metropolitan Statistical Areas (MSAs). The sample is restricted to "in-market" observations, where the bank maintains a physical branch presence in the county. Column (1) presents results for mortgage lending using the full sample. Column (2) reports mortgage lending for the subset of bank-county pairs that also report small business (SMB) lending. Column (3) reports small business lending for the full sample. Both the dependent variables and the deposit channel instrument are normalized by the previous year's total bank-level assets. The sample covers 2009–2023, with all variables winsorized at the 1st and 99th percentiles. All specifications include bank×year, county×year, and bank×county fixed effects. Robust standard errors, two-way clustered at the bank and county levels, are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)
Dependent Variable:	$\Delta\text{Mortgages}_{b,c,t}/\text{Assets}_{b,t-1}$	$\Delta\text{SMB}_{b,c,t}/\text{Assets}_{b,t-1}$	
$Z_{b,c,t}/\text{Assets}_{b,t-1}$	0.25***	0.09***	0.09***
	(0.03)	(0.01)	(0.01)
N	123,141	123,141	127,809
R^2	0.93	0.94	0.93
Bank×Year FE	Yes	Yes	Yes
County×Year FE	Yes	Yes	Yes
Bank×County FE	Yes	Yes	Yes
Sample Period	Full Sample	Full Sample	Full Sample

Table 6: Direct Asset-Swap vs. Deposit Channel: Effects on Employment Growth

This table reports estimates of the effect of the direct asset-swap and deposit channels of QE on county-level employment growth within Metropolitan Statistical Areas (MSAs). The dependent variables in columns (1)–(3) are the quarterly growth rates of total, tradable, and non-tradable employment, following Luck and Zimmermann (2020) and Mian and Sufi (2014). All instruments are scaled by county-level deposits (2008 for QE I–III, 2019 for Pandemic QE) and standardized, so each coefficient is the effect of a one-standard-deviation increase in exposure. The reported coefficients are cumulative dynamic multipliers, summing the contemporaneous and seven-quarter lagged effects. All specifications include county and quarter fixed effects, with standard errors two-way clustered at the county and state-quarter levels in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: QE I–III Period

	(1)	(2)	(3)
Dependent Variable:	$\Delta \ln(\text{Tot Emp})_{c,t}$	$\Delta \ln(\text{Tra Emp})_{c,t}$	$\Delta \ln(\text{Non-Tra Emp})_{c,t}$
$\sum_{j=0}^{7} \text{STD}(Z_{\text{tsy},c,t-j}/\text{Deposits}_{c,2008})$	0.000	0.018***	0.006***
	(0.001)	(0.007)	(0.002)
$\sum_{j=0}^{7} \text{STD}(Z_{\text{mbs},c,t-j}/\text{Deposits}_{c,2008})$	-0.001	-0.017*	0.002
	(0.001)	(0.010)	(0.004)
$\sum_{j=0}^{7} \text{STD}(Z_{c,t-j}/\text{Deposits}_{c,2008})$	0.000	0.001	0.005***
	(0.000)	(0.003)	(0.001)
N	44,439	34,675	43,737
R^2	0.22	0.03	0.36
County FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes

Panel B: Pandemic QE Period

	(1)	(2)	(3)
Dependent Variable:	$\Delta \ln(\text{Tot Emp})_{c,t}$	$\Delta \ln(\text{Tra Emp})_{c,t}$	$\Delta \ln(\text{Non-Tra Emp})_{c,t}$
$\sum_{j=0}^{7} \text{STD}(Z_{\text{tsy},c,t-j}/\text{Deposits}_{c,2019})$	-0.002	-0.006	-0.010
	(0.002)	(0.025)	(0.022)
$\sum_{j=0}^{7} \text{STD}(Z_{\text{mbs},c,t-j}/\text{Deposits}_{c,2019})$	-0.001	0.053	-0.003
	(0.003)	(0.045)	(0.029)
$\sum_{j=0}^{7} \text{STD}(Z_{c,t-j}/\text{Deposits}_{c,2019})$	0.003***	-0.001	0.027***
	(0.001)	(0.010)	(0.010)
N	15,947	12,587	14,395
R^2	0.20	0.06	0.55
County FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes

Table 7: Direct Asset-Swap vs. Deposit Channel: Effects on Local House Prices

This table reports estimates of the effect of the direct asset-swap and deposit channels of QE on local house prices within Metropolitan Statistical Areas (MSAs). Columns (1)–(2) use annual FHFA House Price Index (HPI) data and columns (3)–(4) use quarterly Zillow Home Value Index (ZHVI) data; the dependent variable is the log-difference in the respective index. Columns (1) and (3) cover the QE I–III period (2009–2015), and columns (2) and (4) cover the Pandemic QE period (2020–2022). The primary independent variables are the county-level deposit channel instrument and two instruments capturing the direct asset-swap effect. All instruments are scaled by 2008 (QE I–III) or 2019 (Pandemic QE) total county deposits and then standardized to facilitate interpretation. The annual specifications include the contemporaneous instrument and its one-year lag; the quarterly specifications include the contemporaneous instrument and four quarterly lags. Reported coefficients are the sum of the contemporaneous and lagged terms; annual and quarterly coefficients are not directly comparable, as the quarterly estimates measure per-quarter growth. Columns (1)–(2) include county and year fixed effects (standard errors clustered at the county and state-year levels); columns (3)–(4) include county and quarter fixed effects (standard errors clustered at the county and state-quarter levels). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Annual (FHFA HPI)		Quarterly (Zillow ZHVI)	
	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta \ln(\text{HPI})_{c,t}$	$\Delta \ln(\text{HPI})_{c,t}$	$\Delta \ln(\text{ZHVI})_{c,t}$	$\Delta \ln(\text{ZHVI})_{c,t}$
$\sum_j \text{STD}(Z_{\text{tsy},c,t-j}/\text{Deposits}_{c,t_0})$	0.001	-0.004	0.000	0.000
	(0.001)	(0.004)	(0.000)	(0.001)
$\sum_j \text{STD}(Z_{\text{mbs},c,t-j}/\text{Deposits}_{c,t_0})$	0.004**	0.026**	-0.001*	0.004**
	(0.002)	(0.012)	(0.000)	(0.001)
$\sum_j \text{STD}(Z_{c,t-j}/\text{Deposits}_{c,t_0})$	0.019***	0.011***	0.004***	0.006***
	(0.004)	(0.004)	(0.001)	(0.001)
N	10,449	3,478	37,201	14,135
R^2	0.52	0.80	0.43	0.68
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No
Quarter FE	No	No	Yes	Yes
Sample Period	QE I–III	Pandemic QE	QE I–III	Pandemic QE

Table 8: Direct Asset-Swap vs. Deposit Channel: Effects on Local Mortgage Lending

This table reports estimates of the effect of the direct asset-swap and deposit channels of QE on local home-purchase mortgage lending at the county-year level for counties within Metropolitan Statistical Areas (MSAs). The sample is restricted to home-purchase originations and excludes refinancings. The dependent variables are the log of the annual number of mortgage originations (columns 1 and 3) and the log of their total dollar volume (columns 2 and 4); because HMDA records the annual flow of new originations rather than a stock of loans outstanding, the county fixed effects absorb each county's baseline level of originations. The model is a distributed-lag specification that includes the contemporaneous instruments and their one-year lags; the reported coefficients are cumulative dynamic multipliers, the sum of the contemporaneous and lagged terms. Columns (1)–(2) cover the QE I–III period (2009–2015) and columns (3)–(4) the Pandemic QE period (2020–2022). All instruments are scaled by total county deposits ($t_0 = 2008$ for QE I–III and $t_0 = 2019$ for Pandemic QE) and standardized, so that each coefficient is the approximate percentage difference in originations associated with a one-standard-deviation increase in exposure. All specifications include county and year fixed effects. Robust standard errors, two-way clustered at the county and state-year levels, are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Dependent Variable:	$\ln(\text{Num})_{c,t}$	$\ln(\text{Amt})_{c,t}$	$\ln(\text{Num})_{c,t}$	$\ln(\text{Amt})_{c,t}$
$\sum_{j=0}^{1} \text{STD}(Z_{\text{tsy},c,t-j}/\text{Deposits}_{c,t_0})$	0.008	0.005	-0.015	-0.018
	(0.007)	(0.007)	(0.013)	(0.014)
$\sum_{j=0}^{1} \text{STD}(Z_{\text{mbs},c,t-j}/\text{Deposits}_{c,t_0})$	0.015*	0.023**	0.075	0.093*
	(0.009)	(0.011)	(0.057)	(0.055)
$\sum_{j=0}^{1} \text{STD}(Z_{c,t-j}/\text{Deposits}_{c,t_0})$	0.028***	0.029***	0.047***	0.043***
	(0.009)	(0.010)	(0.013)	(0.014)
N	10,644	10,644	3,548	3,548
R^2	0.99	0.99	0.99	0.99
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Sample Period	QE I-III	QE I-III	Pandemic QE	Pandemic QE

Table 9: The Deposit Channel of QE: Mortgage Approval and Retention

This table reports estimates of the effect of the deposit channel of QE on bank-county mortgage activity within Metropolitan Statistical Areas (MSAs). The dependent variables are the mortgage approval rate (column 1), the log of the annual number of newly retained mortgages (column 2), and the log of their annual dollar volume (column 3); because the retention measures are annual flows, the bank×county fixed effects absorb each bank-county's baseline level of retention. The primary independent variable is the bank-county deposit channel instrument, scaled by total bank assets (2008 for the QE I–III era and 2019 for the Pandemic QE era) and standardized so that each coefficient is interpreted as the effect of a one-standard-deviation increase in exposure. Panel A reports the QE I–III era (2009–2015) and Panel B the Pandemic QE era (2020–2022). All specifications include bank×year, county×year, and bank×county fixed effects, as well as an indicator for physical branch presence in the county. Robust standard errors, two-way clustered at the bank and county levels, are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: QE I–III Period

	(1)	(2)	(3)
Dependent Variable:	$\text{Appr Rate}_{b,c,t}$	$\ln(\text{Ret Num})_{b,c,t}$	$\ln(\text{Ret Amt})_{b,c,t}$
$\text{STD}(Z_{b,c,t}/\text{Assets}_{b,2008})$	0.001**	0.009***	0.008***
	(0.000)	(0.003)	(0.003)
In Branch	-0.035***	0.689***	0.644***
	(0.005)	(0.041)	(0.039)
N	635,157	359,137	359,137
R^2	0.62	0.91	0.88
Bank×Year FE	Yes	Yes	Yes
County×Year FE	Yes	Yes	Yes
Bank×County FE	Yes	Yes	Yes

Panel B: Pandemic QE Period

	(1)	(2)	(3)
Dependent Variable:	$\text{Appr Rate}_{b,c,t}$	$\ln(\text{Ret Num})_{b,c,t}$	$\ln(\text{Ret Amt})_{b,c,t}$
$\text{STD}(Z_{b,c,t}/\text{Assets}_{b,2019})$	0.003***	0.030***	0.035***
	(0.001)	(0.010)	(0.011)
In Branch	-0.030***	0.580***	0.543***
	(0.009)	(0.078)	(0.072)
N	242,722	133,076	133,076
R^2	0.69	0.94	0.92
Bank×Year FE	Yes	Yes	Yes
County×Year FE	Yes	Yes	Yes
Bank×County FE	Yes	Yes	Yes

Table 10: The Deposit Channel of QE: Conforming vs. Jumbo Mortgage Lending

This table reports estimates of the effect of the deposit channel of QE on bank-county mortgage lending within Metropolitan Statistical Areas (MSAs) over the Pandemic QE period, 2020–2022. The dependent variable is the log of the annual number of mortgage originations (Panel A) or the log of their annual dollar volume (Panel B); because these are annual origination flows, the bank×county fixed effects absorb each bank-county's baseline level of originations. Column (1) reports results for all mortgages, and columns (2) and (3) report separate results for conforming and jumbo mortgages, respectively. The primary independent variable is the bank-county deposit channel instrument, scaled by 2019 total bank assets and standardized so that each coefficient is interpreted as the effect of a one-standard-deviation increase in exposure. All specifications include bank×year, county×year, and bank×county fixed effects, as well as an indicator for physical branch presence in the county. Robust standard errors, two-way clustered at the bank and county levels, are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: QE and the Number of Mortgage Loans

	(1)	(2)	(3)
Dependent Variable:	$\ln(\text{Tot Num})_{b,c,t}$	$\ln(\text{Con Num})_{b,c,t}$	$\ln(\text{Jum Num})_{b,c,t}$
$\text{STD}(Z_{b,c,t}/\text{Assets}_{b,2019})$	0.026***	0.024***	0.033***
	(0.010)	(0.010)	(0.008)
In Branch	0.692***	0.685***	0.373***
	(0.077)	(0.074)	(0.093)
N	242,722	234,892	41,595
R^2	0.95	0.95	0.92
Bank×Year FE	Yes	Yes	Yes
County×Year FE	Yes	Yes	Yes
Bank×County FE	Yes	Yes	Yes

Panel B: QE and the Amount of Mortgage Lending

	(1)	(2)	(3)
Dependent Variable:	$\ln(\text{Tot Amt})_{b,c,t}$	$\ln(\text{Con Amt})_{b,c,t}$	$\ln(\text{Jum Amt})_{b,c,t}$
$\text{STD}(Z_{b,c,t}/\text{Assets}_{b,2019})$	0.029***	0.028***	0.033***
	(0.010)	(0.011)	(0.009)
In Branch	0.681***	0.700***	0.391***
	(0.077)	(0.075)	(0.103)
N	242,722	234,892	41,595
R^2	0.93	0.93	0.92
Bank×Year FE	Yes	Yes	Yes
County×Year FE	Yes	Yes	Yes
Bank×County FE	Yes	Yes	Yes

Figure 1: The Asset-Swap Channel versus the Deposit Channel

This figure illustrates the channels through which QE is transmitted to the banking sector. Panel A depicts the textbook asset-swap channel, where banks exchange their securities for central bank reserves. Panel B shows the deposit channel, where banks absorb deposit inflows generated by non-bank clients selling securities to the Fed.

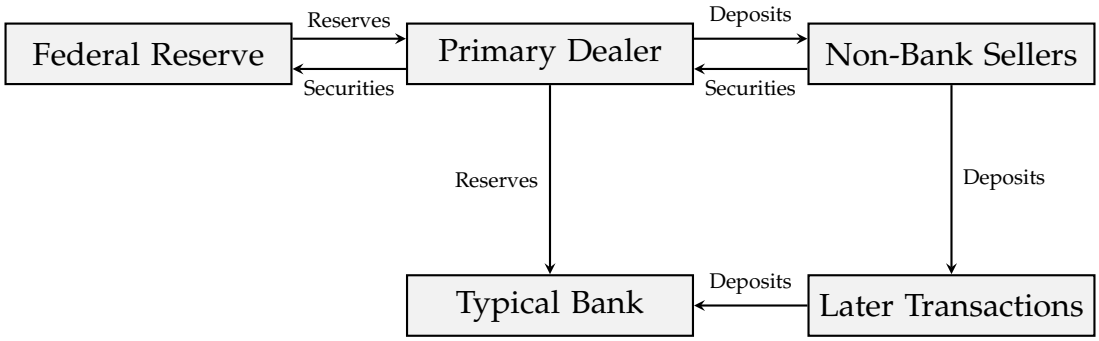
Panel A: Asset-Swap Channel



The Bank's Balance Sheet

Assets		Liabilities
Securities	$(-)$	
Reserves	$(+)$	

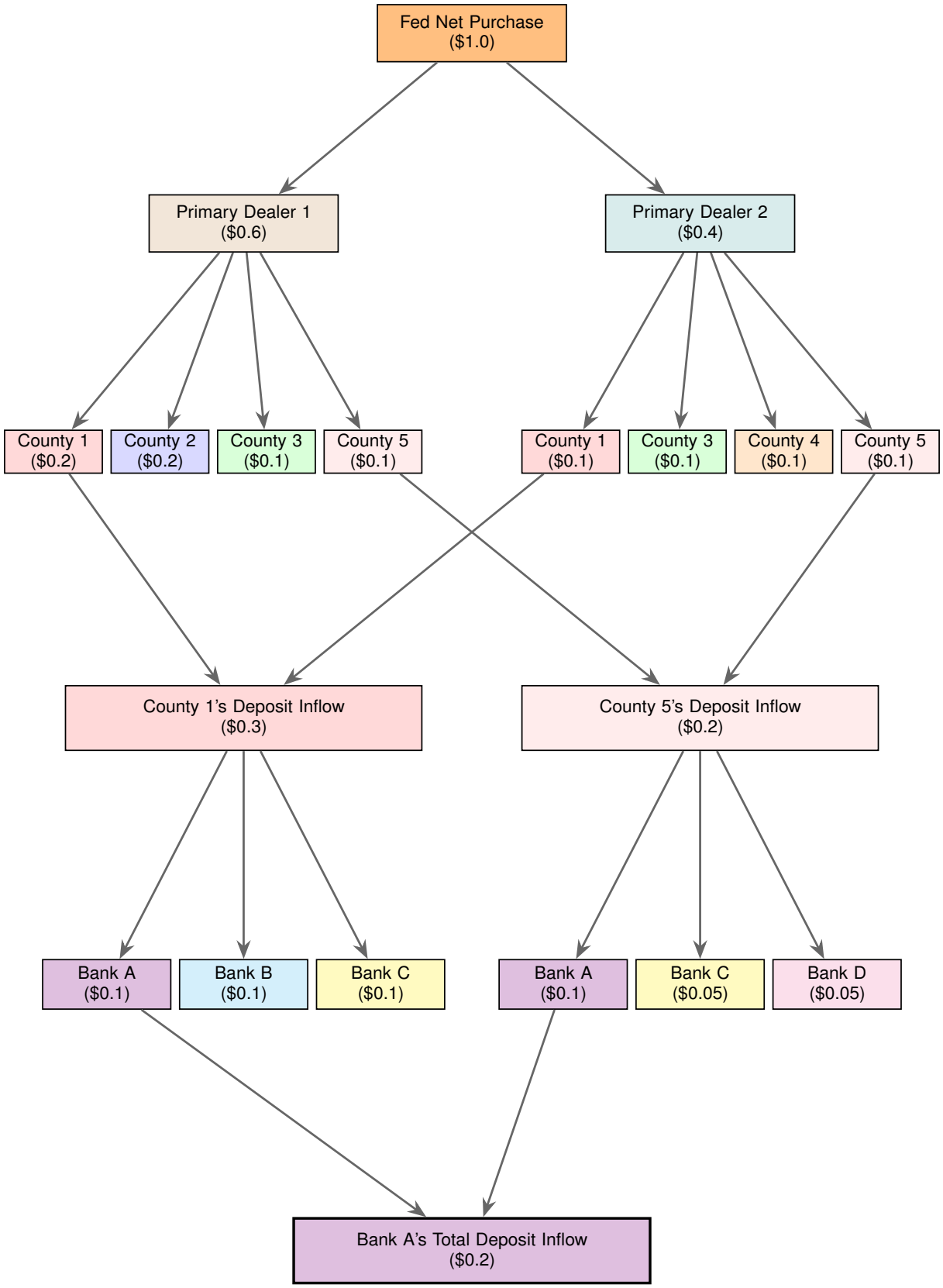
Panel B: Deposit Channel



The Bank's Balance Sheet

Assets		Liabilities	
Reserves	$(+)$	Deposits	$(+)$

Figure 2: Conceptual Framework for Constructing the Deposit Channel Instrument



Appendix

The Deposit Channel of Quantitative Easing

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Table A.1: Variable Definitions

This table provides definitions for the variables and their corresponding data terms used in this study.

Variable	Definition	Data Terms
Panel A: Summary of Deposits (SOD)		
$\text{Deposits}_{c,t}$	Total deposits across all bank branches located in county c as of June 30 of year t.	$\sum_{br \in c} \text{Depsumbr}_{br,t}$
$\text{Deposits}_{b,c,t}$	Total deposits across all branches of bank b located in county c as of June 30 of year t.	$\sum_{br \in bc} \text{Depsumbr}_{br,t}$
Panel B: Call Reports		
$\text{Assets}_{b,t}$	Total assets of bank b at quarter t.	$\max(\text{RCFD2170}_{b,t}, \text{RCON2170}_{b,t})$
$\text{Reserves}_{b,t}$	Total reserves of bank b at quarter t.	$\max(\text{RCFD0090}_{b,t}, \text{RCON0090}_{b,t})$
$\text{Deposits}_{b,t}$	Total deposits of bank b at quarter t.	$\text{RCON2200}_{b,t} + \text{RCFN2200}_{b,t}$
$\text{Loans}_{b,t}$	Total loans of bank b at quarter t.	$\max(\text{RCFD2122}_{b,t}, \text{RCON2122}_{b,t})$ $+ \max(\text{RCFD3814}_{b,t}, \text{RCON3814}_{b,t})$ $+ \max(\text{RCFD3815}_{b,t}, \text{RCON3815}_{b,t})$ $+ \max(\text{RCFDF164}_{b,t}, \text{RCONF164}_{b,t})$ $+ \max(\text{RCFDF165}_{b,t}, \text{RCONF165}_{b,t})$ $+ \max(\text{RCFD6550}_{b,t}, \text{RCON6550}_{b,t})$ $+ \max(\text{RCFD3818}_{b,t}, \text{RCON3818}_{b,t})$ if $t < 2010Q1$ $\max(\text{RCFD2122}_{b,t}, \text{RCON2122}_{b,t})$ $+ \max(\text{RCFD3814}_{b,t}, \text{RCON3814}_{b,t})$ $+ \max(\text{RCFD3815}_{b,t}, \text{RCON3815}_{b,t})$ $+ \max(\text{RCFDF164}_{b,t}, \text{RCONF164}_{b,t})$ $+ \max(\text{RCFDF165}_{b,t}, \text{RCONF165}_{b,t})$ $+ \max(\text{RCFD6550}_{b,t}, \text{RCON6550}_{b,t})$ $+ \max(\text{RCFDJ457}_{b,t}, \text{RCONJ457}_{b,t})$ if $t \geq 2010Q1$
$\text{Mortgages}_{b,t}$	Total mortgages of bank b at quarter t.	$\text{RCON1797}_{b,t} + \text{RCON5367}_{b,t}$ $+\text{RCON5368}_{b,t} + \text{RCON1460}_{b,t}$
$\text{C\&I}_{b,t}$	Total commercial and industrial (C&I) loans of bank b at quarter t.	$\max(\text{RCON1763}_{b,t} + \text{RCON1764}_{b,t}, \text{RCON1766}_{b,t})$ $+ \max(\text{RCFD3818}_{b,t}, \text{RCON3818}_{b,t})$ if $t < 2010Q1$ $\max(\text{RCON1763}_{b,t} + \text{RCON1764}_{b,t}, \text{RCON1766}_{b,t})$ $+ \max(\text{RCFDJ457}_{b,t}, \text{RCONJ457}_{b,t})$ if $t \geq 2010Q1$
$\text{Securities}_{b,t}$	Total securities of bank b at quarter t.	$\max(\text{RCFD1754}_{b,t}, \text{RCON1754}_{b,t})$ $+ \max(\text{RCFD1773}_{b,t}, \text{RCON1773}_{b,t})$

Table A.1 continued from previous page

Variable	Definition	Data Terms
Panel C: Home Mortgage Disclosure Act (HMDA)		
$\text{Num}_{c,t}$	Total number of mortgages originated in county c during the one-year period ending in year t.	$\sum_{m \in c} \mathbb{1}_{m,t}$
$\text{Amt}_{c,t}$	Total dollar amount of mortgages originated in county c during the one-year period ending in year t.	$\sum_{m \in c} \text{Loan_Amount}_{m,t}$
$\Delta\text{Mortgages}_{b,c,t}$	Total dollar amount of mortgages originated by bank b in county c during the one-year period ending in year t.	$\sum_{m \in bc} \text{Loan_Amount}_{m,t}$
$\text{Appr Rate}_{b,c,t}$	The mortgage approval rate of bank b in county c during the one-year period ending in year t.	$\frac{\sum_{m \in bc} \mathbb{1}(\text{Action_Taken} = 1/2)_{m,t}}{\sum_{m \in bc} \mathbb{1}(\text{Action_Taken} = 1/2/3)_{m,t}}$
$\text{Ret Num}_{b,c,t}$	Total number of mortgages originated and retained by bank b in county c during the one-year period ending in year t.	$\sum_{m \in bc} \mathbb{1}(\text{Purchaser_Type} = 0)_{m,t}$
$\text{Ret Amt}_{b,c,t}$	Total dollar amount of mortgages originated and retained by bank b in county c during the one-year period ending in year t.	$\sum_{m \in bc} \mathbb{1}(\text{Purchaser_Type} = 0)_{m,t} \times \text{Loan_Amount}_{m,t}$
$\text{Tot Num}_{b,c,t}$	Total number of mortgages originated by bank b in county c during the one-year period ending in year t.	$\sum_{m \in bc} \mathbb{1}_{m,t}$
$\text{Tot Amt}_{b,c,t}$	Total dollar amount of mortgages originated by bank b in county c during the one-year period ending in year t.	$\sum_{m \in bc} \text{Loan_Amount}_{m,t}$

Table A.1 continued from previous page

Variable	Definition	Data Terms
Panel C: Home Mortgage Disclosure Act (HMDA)		
$\text{Con Num}_{b,c,t}$	Total number of conforming mortgages originated by bank b in county c during the one-year period ending in year t.	$\sum_{m \in bc} \mathbb{1}(\text{Conforming_Loan_Limit} = \text{C})_{m,t}$
$\text{Con Amt}_{b,c,t}$	Total dollar amount of conforming mortgages originated by bank b in county c during the one-year period ending in year t.	$\sum_{m \in bc} \mathbb{1}(\text{Conforming_Loan_Limit} = \text{C})_{m,t} \times \text{Loan_Amount}_{m,t}$
$\text{Jum Num}_{b,c,t}$	Total number of jumbo mortgages originated by bank b in county c during the one-year period ending in year t.	$\sum_{m \in bc} \mathbb{1}(\text{Conforming_Loan_Limit} = \text{NC})_{m,t}$
$\text{Jum Amt}_{b,c,t}$	Total dollar amount of jumbo mortgages originated by bank b in county c during the one-year period ending in year t.	$\sum_{m \in bc} \mathbb{1}(\text{Conforming_Loan_Limit} = \text{NC})_{m,t} \times \text{Loan_Amount}_{m,t}$
Panel D: Community Reinvestment Act (CRA)		
$\Delta\text{SMB}_{b,c,t}$	Total dollar amount of small business loans originated by bank b in county c during the one-year period ending in year t.	$\sum_{m \in bc} \text{Amt_Smb}_{m,t}$
Panel E: Quarterly Census of Employment and Wages (QCEW)		
$\text{Tot Emp}_{c,t}$	Quarterly average of monthly (m) employment in county c during quarter t.	$\frac{1}{3} \sum_{m \in t} \text{Monthi_Emplvl}_{c,m}$
$\text{Tra Emp}_{c,t}$	Quarterly average of monthly employment in the tradable sector for county c in quarter t.	According to Mian and Sufi (2014).
$\text{Non-Tra Emp}_{c,t}$	Quarterly average of monthly employment in the non-tradable sector for county c in quarter t.	According to Mian and Sufi (2014).

Table A.1 continued from previous page

Variable	Definition	Data Terms
Panel F: Federal Housing Finance Agency (FHFA)		
$\text{HPI}_{c,t}$	Annual House Price Index in county c during the one-year period ending in year t.	$\text{HPI}_{c,t}$
Panel G: Zillow		
$\text{ZHVI}_{c,t}$	Quarterly average of monthly (m) Zillow Home Value Index in county c in quarter t.	$\frac{1}{3}\sum_{m \in t} \text{ZHVI}_{c,m}$

Table A.2: Summary Statistics

This table reports summary statistics for the primary variables used in this study. All variables, with the exception of those in logarithmic or standardized form, are winsorized at the 1st and 99th percentiles.

	Mean	SD	Min	Max	N
Panel A: County-Level Variables					
$Z_{c,t}/\text{Deposits}_{c,t-1}$	0.005	0.014	0.000	0.087	46,651
$\Delta\text{Deposits}_{c,t}/\text{Deposits}_{c,t-1}$	0.039	0.077	-0.182	0.314	46,651
$\text{STD}(Z_{\text{tsy},c,t}/\text{Deposits}_{c,2008})$	0.000	1.000	-0.235	7.368	43,996
$\text{STD}(Z_{\text{tsy},c,t}/\text{Deposits}_{c,2019})$	0.000	1.000	-0.242	7.506	21,230
$\text{STD}(Z_{\text{mbs},c,t}/\text{Deposits}_{c,2008})$	0.000	1.000	-0.446	6.314	43,996
$\text{STD}(Z_{\text{mbs},c,t}/\text{Deposits}_{c,2019})$	0.000	1.000	-0.410	6.455	21,230
$\text{STD}(Z_{c,t}/\text{Deposits}_{c,2008})$	0.000	1.000	-0.418	5.778	43,996
$\text{STD}(Z_{c,t}/\text{Deposits}_{c,2019})$	0.000	1.000	-0.471	5.163	21,230
$\Delta\ln(\text{Tot Emp})_{c,t}$	0.004	0.034	-0.108	0.090	26,657
$\Delta\ln(\text{Tra Emp})_{c,t}$	-0.002	0.619	-2.000	2.000	22,264
$\Delta\ln(\text{Non-Tra Emp})_{c,t}$	-0.094	0.321	-1.552	0.677	26,245
$\Delta\ln(\text{HPI})_{c,t}$	0.033	0.062	-0.376	0.321	26,087
$\Delta\ln(\text{ZHVI})_{c,t}$	0.010	0.020	-0.365	0.185	99,815
$\ln(\text{Num})_{c,t}$	6.454	1.614	0.000	12.259	26,651
$\ln(\text{Amt})_{c,t}$	11.604	1.832	2.639	18.520	26,651
Panel B: Bank-Level Variables					
$Z_{b,t}/\text{Assets}_{b,t-1}$	0.002	0.004	0.000	0.026	358,189
$Z_{\text{tsy},b,t}/\text{Assets}_{b,t-1}$	0.000	0.000	0.000	0.000	358,189
$Z_{\text{mbs},b,t}/\text{Assets}_{b,t-1}$	0.002	0.003	0.000	0.016	358,188
$\Delta\text{Reserves}_{b,t}/\text{Assets}_{b,t-1}$	0.001	0.028	-0.088	0.105	98,642
$\Delta\text{Deposits}_{b,t}/\text{Assets}_{b,t-1}$	0.013	0.045	-0.093	0.206	356,272
$\Delta\text{Loans}_{b,t}/\text{Assets}_{b,t-1}$	0.009	0.037	-0.086	0.172	356,272
$\Delta\text{Mortgages}_{b,t}/\text{Assets}_{b,t-1}$	0.007	0.022	-0.049	0.108	356,272
$\Delta\text{C\&I}_{b,t}/\text{Assets}_{b,t-1}$	0.001	0.016	-0.052	0.072	356,272
$\Delta\text{Securities}_{b,t}/\text{Assets}_{b,t-1}$	0.003	0.023	-0.064	0.096	356,272
Panel C: Bank-County Level Variables					
$Z_{b,c,t}/\text{Assets}_{b,t-1}$	0.001	0.005	0.000	0.037	351,625
$\Delta\text{Deposits}_{b,c,t}/\text{Assets}_{b,t-1}$	0.011	0.037	-0.088	0.192	337,875
$\Delta\text{Mortgages}_{b,c,t}/\text{Assets}_{b,t-1}$	0.001	0.002	0.000	0.016	618,605
$\Delta\text{SMB}_{b,c,t}/\text{Assets}_{b,t-1}$	0.000	0.001	0.000	0.004	1,071,419
$\text{STD}(Z_{b,c,t}/\text{Assets}_{b,2008})$	0.000	1.000	-0.117	9.377	2,880,612
$\text{STD}(Z_{b,c,t}/\text{Assets}_{b,2019})$	0.000	1.000	-0.106	9.716	757,965
$\text{Appr Rate}_{b,c,t}$	0.887	0.183	0.005	1.000	2,677,398
$\ln(\text{Ret Num})_{b,c,t}$	1.159	1.356	0.000	8.981	1,132,234
$\ln(\text{Ret Amt})_{b,c,t}$	6.461	1.606	0.000	15.695	1,132,206
$\ln(\text{Tot Num})_{b,c,t}$	1.387	1.538	0.000	10.202	2,677,398
$\ln(\text{Tot Amt})_{b,c,t}$	6.692	1.704	0.000	16.347	2,677,370
$\ln(\text{Con Num})_{b,c,t}$	1.380	1.532	0.000	10.136	2,630,204
$\ln(\text{Con Amt})_{b,c,t}$	6.628	1.671	0.000	16.219	2,630,204
$\ln(\text{Jum Num})_{b,c,t}$	0.839	1.133	0.000	8.401	383,892
$\ln(\text{Jum Amt})_{b,c,t}$	7.457	1.232	6.035	15.619	383,839

Table A.3: Physical Presence of Bank-Affiliated U.S. Primary Dealers

This table identifies the counties where U.S. primary dealers with commercial bank subsidiaries maintained a physical presence as of 2008. The Business Share for each dealer-county pair is defined as the ratio of the dealer's deposits within a specific county to its total nationwide deposits as of June 30, 2008.

Primary Dealer	Counties Covered
Bank of America	Abbeville, SC; Ada, ID; Adams, IL; Adams, WA; Aiken, SC; Alachua, FL; Alamance, NC; Alameda, CA; Albany, NY; Albemarle, VA; Alexandria City, VA; Allegan, MI; Amador, CA; Anderson, KS; Anderson, SC; Anderson, TN; Andrews, TX; Androscoggin, ME; Anne Arundel, MD; Arlington, VA; Asotin, WA; Atchison, KS; Atlantic, NJ; Audrain, MO; Avery, NC; Baker, OR; Baltimore City, MD; Baltimore County, MD; Bannock, ID; Barnstable, MA; Barrow, GA; Barry, MO; Bartow, GA; Bates, MO; Baxter, AR; Bay, MI; Beaufort, NC; Beaufort, SC; Ben Hill, GA; Benton, AR; Benton, OR; Benton, WA; Bergen, NJ; Berkeley, SC; Berks, PA; Bernalillo, NM; Berrien, MI; Bexar, TX; Bibb, GA; Bladen, NC; Blaine, ID; Bonner, ID; Bonneville, ID; Boone, AR; Boone, MO; Bosque, TX; Botetourt, VA; Brazos, TX; Brevard, FL; Bristol, MA; Bristol, RI; Bronx, NY; Broward, FL; Brown, TX; Brunswick, NC; Brunswick, VA; Buckingham, VA; Bucks, PA; Buncombe, NC; Burke, NC; Burlington, NJ; Butte, CA; Butts, GA; Cabarrus, NC; Calaveras, CA; Caldwell, NC; Calhoun, FL; Calhoun, MI; Calvert, MD; Camden, NJ; Cameron, TX; Campbell, VA; Canyon, ID; Cape Girardeau, MO; Cape May, NJ; Carbon, PA; Caroline, MD; Carroll, GA; Carroll, MD; Carroll, VA; Carson City, NV; Carteret, NC; Cass, MI; Cass, MO; Catawba, NC; Catoosa, GA; Cayuga, NY; Cerro Gordo, IA; Charles, MD; Charleston, SC; Charlotte, FL; Charlottesville City, VA; Chatham, GA; Chatham, NC; Chaves, NM; Cheatham, TN; Chelan, WA; Cherokee, GA; Cherokee, SC; Chesapeake City, VA; Cheshire, NH; Chester, PA; Chesterfield, VA; Choctaw, OK; Churchill, NV; Citrus, FL; Clackamas, OR; Clallam, WA; Clark, NV; Clark, WA; Clarke, GA; Clatsop, OR; Clay, FL; Clay, MO; Clayton, GA; Clayton, IA; Cleveland, OK; Clinton, IL; Clinton, MI; Coal, OK; Cobb, GA; Cochise, AZ; Coconino, AZ; Cole, MO; Coles, IL; Colfax, NM; Collier, FL; Collin, TX; Colonial Heights City, VA; Colquitt, GA; Columbia, FL; Columbia, GA; Columbia, NY; Columbia, OR; Columbia, WA; Colusa, CA; Comal, TX; Contra Costa, CA; Cook, IL; Coos, OR; Cortland, NY; Coryell, TX; Cotton, OK; Coweta, GA; Craighead, AR; Craven, NC; Crawford, KS; Crawford, MO; Cumberland, ME; Cumberland, NC; Cumberland, NJ; Curry, NM

Table A.3 continued from previous page

Primary Dealer	**Counties Covered**
Bank of America	Dallas, TX; Danville City, VA; Dare, NC; Darlington, SC; Davidson, NC; Davidson, TN; Davie, NC; DeKalb, GA; DeSoto, FL; Del Norte, CA; Delaware, PA; Denton, TX; Deschutes, OR; District of Columbia, DC; Dixie, FL; Dona Ana, NM; Dorchester, MD; Dorchester, SC; Dougherty, GA; Douglas, GA; Douglas, KS; Douglas, NV; Douglas, OR; Douglas, WA; DuPage, IL; Dubuque, IA; Duplin, NC; Durham, NC; Dutchess, NY; Duval, FL; Eaton, MI; El Dorado, CA; El Paso, TX; Elbert, GA; Elkhart, IN; Elko, NV; Ellis, KS; Ellis, TX; Erath, TX; Erie, NY; Escambia, FL; Essex, MA; Essex, NJ; Fairfax City, VA; Fairfax, VA; Fairfield, CT; Falls Church City, VA; Falls, TX; Faulkner, AR; Fayette, GA; Ferry, WA; Finney, KS; Flagler, FL; Florence, SC; Ford, KS; Forsyth, GA; Forsyth, NC; Fort Bend, TX; Franklin City, VA; Franklin, IL; Franklin, MA; Franklin, MO; Franklin, VA; Frederick, MD; Fredericksburg City, VA; Fresno, CA; Fulton, GA; Fulton, NY; Gadsden, FL; Galax City, VA; Galveston, TX; Garfield, OK; Garfield, WA; Garland, AR; Gasconade, MO; Gaston, NC; Geary, KS; Genesee, MI; Genesee, NY; Georgetown, SC; Gibson, TN; Gila, AZ; Glenn, CA; Gloucester, NJ; Glynn, GA; Grafton, NH; Graham, AZ; Grant, NM; Grant, WA; Gratiot, MI; Gray, TX; Grays Harbor, WA; Grayson, TX; Greene, MO; Greene, NY; Greene, VA; Greenville, SC; Greenwood, SC; Guadalupe, TX; Guilford, NC; Gulf, FL; Gwinnett, GA; Halifax, VA; Hall, GA; Hamilton, FL; Hamilton, TN; Hampden, MA; Hampshire, MA; Hanover, VA; Harford, MD; Harris, TX; Harrisonburg City, VA; Hartford, CT; Harvey, KS; Hays, TX; Henderson, NC; Hennepin, MN; Henrico, VA; Henry, GA; Henry, MO; Herkimer, NY; Hernando, FL; Hidalgo, TX; Highlands, FL; Hill, TX; Hillsborough, FL; Hillsborough, NH; Hood River, OR; Hopewell City, VA; Horry, SC; Houston, GA; Howard, MD; Hudson, NJ; Humboldt, CA; Humboldt, NV; Humphreys, TN; Hunt, TX; Hunterdon, NJ; Idaho, ID; Imperial, CA; Indian River, FL; Ingham, MI; Inyo, CA; Iredell, NC; Irwin, GA; Island, WA; Isle of Wight, VA; Jackson, MI; Jackson, MO; Jackson, OK; Jackson, OR; James City, VA; Jasper, MO; Jasper, SC; Jefferson, AR; Jefferson, IL; Jefferson, MO; Jefferson, TX; Jim Wells, TX; Johnson, KS; Johnston, NC; Josephine, OR; Kalamazoo, MI; Kane, IL; Kaufman, TX; Kendall, TX; Kennebec, ME; Kent, MI; Kent, RI; Kern, CA; Kerr, TX; Kershaw, SC

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Primary Dealer	Counties Covered
Bank of America	King, WA; Kings, CA; Kings, NY; Kitsap, WA; Kittitas, WA; Klamath, OR; Knox, ME; Knox, TN; Kossuth, IA; La Paz, AZ; Lackawanna, PA; Laclede, MO; Lafayette, MO; Lake, CA; Lake, FL; Lake, IL; Lamar, TX; Lancaster, PA; Lane, OR; Lassen, CA; Latah, ID; Lauderdale, TN; Laurens, GA; Lawrence, AR; Lawrence, MO; Lea, NM; Lee, FL; Lehigh, PA; Lenawee, MI; Lenoir, NC; Leon, FL; Lewis, ID; Lewis, WA; Lexington City, VA; Lexington, SC; Liberty, GA; Lincoln, MO; Lincoln, NC; Lincoln, NV; Lincoln, OR; Lincoln, WA; Linn, OR; Litchfield, CT; Livingston, MI; Los Angeles, CA; Loudoun, VA; Lowndes, GA; Lubbock, TX; Lunenburg, VA; Luzerne, PA; Lynchburg City, VA; Lyon, KS; Lyon, NV; Macomb, MI; Macon, IL; Macoupin, IL; Madera, CA; Madison, FL; Madison, IL; Malheur, OR; Manassas City, VA; Manatee, FL; Maricopa, AZ; Marin, CA; Marion, FL; Marion, OR; Marshall, TN; Martin, FL; Martinsville City, VA; Mason, WA; Mathews, VA; Maury, TN; McHenry, IL; McKinley, NM; McLennan, TX; McPherson, KS; Mecklenburg, NC; Mecklenburg, VA; Mendocino, CA; Merced, CA; Mercer, NJ; Merrimack, NH; Miami-Dade, FL; Middlesex, CT; Middlesex, MA; Middlesex, NJ; Midland, MI; Midland, TX; Mineral, NV; Mitchell, NC; Modoc, CA; Mohave, AZ; Monmouth, NJ; Mono, CA; Monroe, FL; Monroe, GA; Monroe, MI; Monroe, NY; Monterey, CA; Montgomery, IL; Montgomery, KS; Montgomery, MD; Montgomery, NY; Montgomery, PA; Montgomery, TN; Montgomery, TX; Montgomery, VA; Moore, NC; Moore, TX; Morris, NJ; Multnomah, OR; Murray, GA; Muskegon, MI; Muskogee, OK; Nacogdoches, TX; Nantucket, MA; Napa, CA; Nassau, FL; Nassau, NY; Navajo, AZ; Nelson, VA; Nevada, CA; New Castle, DE; New Hanover, NC; New Haven, CT; New London, CT; New York, NY; Newberry, SC; Newport News City, VA; Newport, RI; Newton, GA; Newton, MO; Niagara, NY; Norfolk City, VA; Norfolk, MA; Northampton, PA; Norton City, VA; Nueces, TX; Nye, NV; Oakland, MI; Ocean, NJ; Oconee, GA; Oconee, SC; Okaloosa, FL; Okanogan, WA; Okeechobee, FL; Oklahoma, OK; Oneida, NY; Onondaga, NY; Onslow, NC; Ontario, NY; Orange, CA; Orange, FL; Orange, NC; Orange, NY; Orange, VA; Orangeburg, SC; Orleans, NY; Osceola, FL; Oswego, NY; Otsego, NY; Ottawa, MI; Pacific, WA; Palm Beach, FL; Palo Pinto, TX; Pasco, FL; Passaic, NJ; Paulding, GA; Payette, ID; Peach, GA; Pend Oreille, WA; Penobscot, ME; Petersburg City, VA; Phelps, MO; Philadelphia, PA; Pickens, SC; Pierce, WA; Pike, GA; Pima, AZ; Pinal, AZ; Pinellas, FL; Pitt, NC; Placer, CA; Platte, MO; Plumas, CA; Plymouth, MA; Polk, FL; Polk, IA; Polk, OR; Pope, AR; Potter, TX

Table A.3 continued from previous page

Primary Dealer	**Counties Covered**
Bank of America	Prince Edward, VA; Prince George's, MD; Prince William, VA; Providence, RI; Pulaski, AR; Pulaski, MO; Pulaski, VA; Putnam, FL; Queen Anne's, MD; Queens, NY; Randall, TX; Randolph, NC; Ray, MO; Reno, KS; Rensselaer, NY; Reynolds, MO; Richland, SC; Richmond City, VA; Richmond, GA; Richmond, NY; Rio Arriba, NM; Riverside, CA; Roane, TN; Roanoke City, VA; Roanoke, VA; Robertson, TN; Robeson, NC; Rockdale, GA; Rockingham, NC; Rockingham, NH; Rockingham, VA; Rockwall, TX; Rogers, OK; Rowan, NC; Rusk, TX; Rutherford, NC; Rutherford, TN; Sacramento, CA; Sagadahoc, ME; Salem City, VA; Saline, KS; San Benito, CA; San Bernardino, CA; San Diego, CA; San Francisco, CA; San Jacinto, TX; San Joaquin, CA; San Juan, NM; San Luis Obispo, CA; San Mateo, CA; Sandoval, NM; Sanilac, MI; Santa Barbara, CA; Santa Clara, CA; Santa Cruz, AZ; Santa Cruz, CA; Santa Fe, NM; Santa Rosa, FL; Sarasota, FL; Saratoga, NY; Schenectady, NY; Schoharie, NY; Schuylkill, PA; Scott, VA; Scurry, TX; Seminole, FL; Seneca, NY; Shasta, CA; Shawnee, KS; Shelby, TN; Shenandoah, VA; Shoshone, ID; Siskiyou, CA; Skagit, WA; Skamania, WA; Smith, TN; Snohomish, WA; Socorro, NM; Solano, CA; Somerset, NJ; Sonoma, CA; Spalding, GA; Spartanburg, SC; Spokane, WA; Spotsylvania, VA; St. Charles, MO; St. Clair, IL; St. Clair, MI; St. Francois, MO; St. Johns, FL; St. Joseph, IN; St. Joseph, MI; St. Louis City, MO; St. Louis County, MO; St. Lucie, FL; St. Mary's, MD; Stafford, VA; Stanislaus, CA; Staunton City, VA; Stevens, WA; Stewart, TN; Stone, MO; Storey, NV; Strafford, NH; Suffolk, MA; Suffolk, NY; Sullivan, NY; Sumner, KS; Sumner, TN; Sumter, FL; Sumter, SC; Sutter, CA; Suwannee, FL; Taney, MO; Tarrant, TX; Taylor, TX; Tehama, CA; Thurston, WA; Titus, TX; Tolland, CT; Tom Green, TX; Tompkins, NY; Towns, GA; Travis, TX; Troup, GA; Tulare, CA; Tulsa, OK; Tuolumne, CA; Twin Falls, ID; Ulster, NY; Umatilla, OR; Union, NC; Union, NJ; Union, OR; Upshur, TX; Vance, NC; Ventura, CA; Vernon, MO; Victoria, TX; Virginia Beach City, VA; Volusia, FL; Wahkiakum, WA; Wake, NC; Walker, TX; Walla Walla, WA; Wallowa, OR; Walton, FL; Warren, NJ; Warren, NY; Washington, AR; Washington, IA; Washington, MD; Washington, NC; Washington, OR; Washington, RI; Washington, TN; Washington, VA; Washoe, NV; Washtenaw, MI; Watauga, NC; Wayne, GA; Wayne, MI; Wayne, NC; Wayne, NY; Waynesboro City, VA; Webb, TX; Westchester, NY; Westmoreland, VA; Whatcom, WA; White Pine, NV; Whitfield, GA; Whitman, WA; Wichita, TX; Wicomico, MD; Will, IL; Williamson, TN; Williamson, TX; Wilson, NC; Wilson, TN; Windham, CT; Winnebago, IL; Wise, VA; Woodbury, IA; Worcester, MA; Worcester, MD; Wyandotte, KS; Yamhill, OR; Yavapai, AZ; Yolo, CA; York, ME; York, SC; Yuba, CA; Yuma, AZ

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Primary Dealer	Counties Covered
JPMorgan Chase	Adams, CO; Albany, NY; Allen, IN; Allen, OH; Arapahoe, CO; Ascension, LA; Ashland, OH; Athens, OH; Auglaize, OH; Austin, TX; Beauregard, LA; Belmont, OH; Bergen, NJ; Berrien, MI; Bexar, TX; Boone, IN; Boone, KY; Bosque, TX; Bossier, LA; Boulder, CO; Brazoria, TX; Brazos, TX; Bronx, NY; Broome, NY; Brown, WI; Bullitt, KY; Butler, OH; Cabell, WV; Caddo, LA; Calcasieu, LA; Calhoun, MI; Cameron, TX; Campbell, KY; Champaign, IL; Charlevoix, MI; Chemung, NY; Choctaw, OK; Clark, IN; Clermont, OH; Clinton, MI; Coal, OK; Cochise, AZ; Coconino, AZ; Collier, FL; Collin, TX; Columbiana, OH; Comal, TX; Cook, IL; Cooke, TX; Coshocton, OH; Cuyahoga, OH; Dallas, TX; Dane, WI; Darke, OH; Daviess, KY; Davis, UT; Delaware, IN; Delaware, OH; Denton, TX; Denver, CO; District of Columbia, DC; Dodge, WI; Douglas, CO; DuPage, IL; Dutchess, NY; East Baton Rouge, LA; Eaton, MI; El Paso, CO; El Paso, TX; Elkhart, IN; Ellis, TX; Emmet, MI; Erie, NY; Erie, OH; Essex, NJ; Fairfield, CT; Fayette, KY; Fayette, WV; Floyd, IN; Fort Bend, TX; Franklin, OH; Fulton, GA; Galveston, TX; Geauga, OH; Genesee, MI; Gila, AZ; Goliad, TX; Graham, AZ; Grand Traverse, MI; Grayson, TX; Green, WI; Greene, OH; Gregg, TX; Grundy, IL; Guadalupe, TX; Hamilton, IN; Hamilton, OH; Hancock, IN; Hardin, KY; Harris, TX; Harrison, IN; Harrison, TX; Harrison, WV; Hays, TX; Hendricks, IN; Henry, OH; Hidalgo, TX; Hillsborough, FL; Hockley, TX; Hood, TX; Hopkins, TX; Hudson, NJ; Hunt, TX; Iberville, LA; Ingham, MI; Ionia, MI; Jefferson, CO; Jefferson, KY; Jefferson, LA; Jefferson, OH; Jefferson, TX; Jefferson, WI; Jessamine, KY; Johnson, IN; Kalamazoo, MI; Kalkaska, MI; Kanawha, WV; Kane, IL; Kankakee, IL; Kaufman, TX; Kenosha, WI; Kent, MI; Kenton, KY; King, WA; Kings, NY; La Crosse, WI; La Paz, AZ; LaPorte, IN; LaSalle, IL; Lafayette, LA; Lafourche, LA; Lake, IL; Lake, IN; Lake, OH; Lapeer, MI; Larimer, CO; Lawrence, IN; Lee, FL; Leelanau, MI; Licking, OH; Lincoln, LA; Livingston, MI; Logan, WV; Lorain, OH; Los Angeles, CA; Macomb, MI; Madison, KY; Madison, NY; Madison, OH; Mahoning, OH; Maricopa, AZ; Marion, IN; Marion, OH; McHenry, IL; McLean, IL; McLennan, TX; Medina, OH; Mercer, NJ; Mercer, OH; Miami, OH; Miami-Dade, FL; Middlesex, NJ; Midland, TX; Milwaukee, WI; Mohave, AZ; Monmouth, NJ; Monroe, IN; Monroe, NY; Montcalm, MI; Montgomery, IN; Montgomery, OH; Montgomery, TX; Morgan, IN; Morris, NJ; Muskegon, MI; Muskingum, OH; Nacogdoches, TX; Nassau, NY; Navajo, AZ; New Castle, DE; New Haven, CT; New York, NY; Noble, IN; Oakland, MI; Ocean, NJ; Ohio, WV; Oklahoma, OK; Oldham, KY; Oneida, NY; Onondaga, NY; Ontario, NY; Orange, NY; Orange, TX; Orleans, LA; Oswego, NY; Ottawa, MI; Ottawa, OH; Ouachita, LA; Outagamie, WI; Ozaukee, WI

Table A.3 continued from previous page

Primary Dealer	**Counties Covered**
JPMorgan Chase	Palm Beach, FL; Passaic, NJ; Peoria, IL; Philadelphia, PA; Pickaway, OH; Pima, AZ; Pinal, AZ; Portage, OH; Portage, WI; Porter, IN; Potter, TX; Preble, OH; Putnam, NY; Putnam, WV; Queens, NY; Racine, WI; Raleigh, WV; Rapides, LA; Richland, OH; Richmond, NY; Rock Island, IL; Rock, WI; Rockland, NY; Rockwall, TX; Ross, OH; Saginaw, MI; Salt Lake, UT; San Francisco, CA; Sangamon, IL; Santa Cruz, AZ; Sarasota, FL; Scott, IN; Scurry, TX; Shelby, IN; Shelby, KY; Shiawassee, MI; Somerset, NJ; St. Bernard, LA; St. Charles, LA; St. Clair, MI; St. Joseph, IN; St. Joseph, MI; St. Mary, LA; St. Tammany, LA; Stark, OH; Suffolk, MA; Suffolk, NY; Sullivan, NY; Summit, OH; Summit, UT; Sussex, NJ; Tarrant, TX; Taylor, TX; Terrebonne, LA; Tippecanoe, IN; Travis, TX; Trumbull, OH; Tulsa, OK; Tuscarawas, OH; Tyler, TX; Ulster, NY; Union, NJ; Union, OH; Van Buren, MI; Vermilion, LA; Walworth, WI; Ward, TX; Warren, KY; Warren, NJ; Warren, OH; Washington, LA; Washington, OH; Washington, WI; Washtenaw, MI; Waukesha, WI; Waupaca, WI; Wayne, MI; Wayne, NY; Wayne, OH; Webb, TX; Weber, UT; Weld, CO; West Baton Rouge, LA; Westchester, NY; White, IN; Whitley, IN; Wichita, TX; Will, IL; Williamson, TX; Winnebago, IL; Winnebago, WI; Wood, OH; Wood, WV; Wyoming, WV; Yavapai, AZ; Yuma, AZ

Table A.3 continued from previous page

Primary Dealer	**Counties Covered**
Wells Fargo	Ada, ID; Adams, CO; Adams, IN; Adams, NE; Alameda, CA; Albany, WY; Allen, IN; Amador, CA; Anchorage, AK; Angelina, TX; Anoka, MN; Apache, AZ; Aransas, TX; Arapahoe, CO; Archuleta, CO; Asotin, WA; Atascosa, TX; Austin, TX; Baker, OR; Bandera, TX; Bannock, ID; Barnes, ND; Bastrop, TX; Beadle, SD; Beaver, UT; Beaverhead, MT; Becker, MN; Beltrami, MN; Benton, IA; Benton, MN; Benton, OR; Benton, WA; Bernalillo, NM; Bethel, AK; Bexar, TX; Big Stone, MN; Black Hawk, IA; Blaine, ID; Blaine, MT; Blue Earth, MN; Bonner, ID; Bonneville, ID; Boulder, CO; Boundary, ID; Box Butte, NE; Box Elder, UT; Brazoria, TX; Brazos, TX; Bristol Bay, AK; Brookings, SD; Broomfield, CO; Brown, MN; Brown, SD; Brown, WI; Brule, SD; Buffalo, NE; Burleigh, ND; Butte, CA; Butte, SD; Cache, UT; Calaveras, CA; Caldwell, TX; Calhoun, TX; Cameron, TX; Campbell, WY; Canyon, ID; Carbon, MT; Carbon, UT; Carlton, MN; Carson City, NV; Cascade, MT; Cass, IA; Cass, ND; Cavalier, ND; Cerro Gordo, IA; Chaves, NM; Chelan, WA; Cherry, NE; Chouteau, MT; Churchill, NV; Cibola, NM; Clackamas, OR; Clallam, WA; Clark, NV; Clatsop, OR; Clay, MN; Clay, SD; Clay, TX; Cochise, AZ; Coconino, AZ; Codington, SD; Colfax, NM; Collin, TX; Columbia, OR; Colusa, CA; Comal, TX; Contra Costa, CA; Cook, IL; Coos, OR; Cowlitz, WA; Crawford, IA; Crook, OR; Crow Wing, MN; Cuming, NE; Curry, NM; Custer, CO; Dakota, MN; Dallas, IA; Dallas, TX; Dane, WI; Davis, UT; Davison, SD; DeKalb, IN; DeWitt, TX; Deer Lodge, MT; Delta, CO; Delta, MI; Denton, TX; Denver, CO; Deschutes, OR; Deuel, SD; Dickinson, MI; Dillingham, AK; Dodge, MN; Dona Ana, NM; Douglas, CO; Douglas, MN; Douglas, NE; Douglas, NV; Douglas, OR; Duchesne, UT; Eagle, CO; El Dorado, CA; El Paso, CO; El Paso, TX; Elbert, CO; Elko, NV; Ellis, TX; Emery, UT; Erath, TX; Fairbanks North Star, AK; Fall River, SD; Fallon, MT; Faribault, MN; Fayette, TX; Fergus, MT; Fillmore, MN; Flathead, MT; Fond du Lac, WI; Fort Bend, TX; Freeborn, MN; Freestone, TX; Fremont, CO; Fremont, WY; Fresno, CA; Gallatin, MT; Galveston, TX; Garfield, CO; Garfield, UT; Gibson, IN; Gila, AZ; Gillespie, TX; Gilpin, CO; Gogebic, MI; Gonzales, TX; Grand Forks, ND; Grand Traverse, MI; Grand, CO; Grand, UT; Grant, NM; Grant, SD; Grays Harbor, WA; Green, WI; Greene, IA; Grimes, TX

Table A.3 continued from previous page

Primary Dealer	Counties Covered
Wells Fargo	Guadalupe, NM; Guadalupe, TX; Hale, TX; Hall, NE; Harris, TX; Hartley, TX; Hays, TX; Hennepin, MN; Hidalgo, TX; Hill, MT; Hill, TX; Hood River, OR; Houghton, MI; Howard, TX; Hughes, SD; Humboldt, CA; Humboldt, NV; Huntington, IN; Hutchinson, TX; Idaho, ID; Imperial, CA; Iowa, WI; Iron, MI; Iron, UT; Isanti, MN; Island, WA; Itasca, MN; Jackson, OR; Jackson, TX; Jefferson, CO; Jefferson, OR; Jefferson, TX; Jim Hogg, TX; Jim Wells, TX; Jones, IA; Jones, TX; Josephine, OR; Juab, UT; Juneau, AK; Kandiyohi, MN; Kaufman, TX; Kenai Peninsula, AK; Kendall, TX; Kent, MI; Kern, CA; Kerr, TX; Ketchikan Gateway, AK; King, WA; Kings, CA; Kingsbury, SD; Kitsap, WA; Kittitas, WA; Klamath, OR; Kleberg, TX; Kodiak Island, AK; Koochiching, MN; La Crosse, WI; La Paz, AZ; La Plata, CO; LaGrange, IN; LaPorte, IN; Lake of the Woods, MN; Lake, CA; Lake, MN; Lake, SD; Lancaster, NE; Lander, NV; Lane, OR; Laramie, WY; Larimer, CO; Latah, ID; Lavaca, TX; Lawrence, SD; Lea, NM; Lee, TX; Leon, TX; Lewis and Clark, MT; Lewis, ID; Lewis, WA; Liberty, MT; Lincoln, NE; Lincoln, NM; Lincoln, OR; Lincoln, SD; Lincoln, WY; Linn, IA; Linn, OR; Logan, CO; Los Alamos, NM; Los Angeles, CA; Lubbock, TX; Luna, NM; Lyman, SD; Lyon, MN; Lyon, NV; Madera, CA; Madison, NE; Malheur, OR; Manitowoc, WI; Marathon, WI; Maricopa, AZ; Marin, CA; Marinette, WI; Marion, IA; Marion, IN; Marion, OR; Marquette, MI; Marshall, IA; Marshall, SD; Matagorda, TX; Matanuska-Susitna, AK; McCone, MT; McCook, SD; McKinley, NM; McLennan, TX; McLeod, MN; McMullen, TX; McPherson, SD; Meeker, MN; Mendocino, CA; Menominee, MI; Merced, CA; Merrick, NE; Mesa, CO; Miami, IN; Midland, TX; Millard, UT; Milwaukee, WI; Minnehaha, SD; Missoula, MT; Mohave, AZ; Monroe, WI; Montague, TX; Monterey, CA; Montgomery, TX; Montrose, CO; Morgan, UT; Morton, ND; Mower, MN; Multnomah, OR; Musselshell, MT; Napa, CA; Natrona, WY; Navajo, AZ; Nevada, CA; Nicollet, MN; Nobles, MN; Nome, AK; North Slope, AK; Northwest Arctic, AK; Nueces, TX; Nye, NV; Okanogan, WA; Olmsted, MN; Orange, CA; Orange, TX; Otero, NM; Otter Tail, MN; Outagamie, WI; Ozaukee, WI; Park, MT; Parker, TX; Payette, ID; Pembina, ND; Pennington, MN; Pennington, SD; Pershing, NV; Petroleum, MT; Pierce, WA; Pima, AZ; Pinal, AZ; Pitkin, CO; Piute, UT; Placer, CA; Platte, WY

Table A.3 continued from previous page

Primary Dealer	Counties Covered
Wells Fargo	Polk, IA; Polk, OR; Pottawattamie, IA; Potter, TX; Powell, MT; Price, WI; Prince of Wales-Hyder, AK; Pueblo, CO; Quay, NM; Racine, WI; Ramsey, MN; Randall, TX; Ravalli, MT; Red Willow, NE; Rice, MN; Rich, UT; Richland, ND; Rio Arriba, NM; Riverside, CA; Roberts, SD; Robertson, TX; Rock Island, IL; Roosevelt, MT; Roosevelt, NM; Roseau, MN; Routt, CO; Rush, IN; Sacramento, CA; Salt Lake, UT; San Benito, CA; San Bernardino, CA; San Diego, CA; San Francisco, CA; San Jacinto, TX; San Joaquin, CA; San Juan, NM; San Juan, WA; San Luis Obispo, CA; San Mateo, CA; San Miguel, CO; San Miguel, NM; Sandoval, NM; Sanpete, UT; Santa Barbara, CA; Santa Clara, CA; Santa Cruz, AZ; Santa Cruz, CA; Santa Fe, NM; Sarpy, NE; Sauk, WI; Scott, IA; Scott, MN; Sevier, UT; Shasta, CA; Sheboygan, WI; Sherburne, MN; Shoshone, ID; Sibley, MN; Sierra, CA; Silver Bow, MT; Sitka, AK; Skagit, WA; Skagway, AK; Skamania, WA; Smith, TX; Snohomish, WA; Socorro, NM; Solano, CA; Sonoma, CA; Southeast Fairbanks, AK; Spink, SD; Spokane, WA; St. Croix, WI; St. Joseph, IN; St. Louis, MN; Stanislaus, CA; Stark, ND; Story, IA; Stutsman, ND; Sullivan, IN; Summit, CO; Summit, UT; Sutter, CA; Sweetwater, WY; Taos, NM; Tarrant, TX; Tehama, CA; Teller, CO; Teton, MT; Teton, WY; Tillamook, OR; Todd, MN; Todd, SD; Tom Green, TX; Tooele, UT; Trail, ND; Traverse, MN; Travis, TX; Tulare, CA; Tuolumne, CA; Turner, SD; Tuscola, MI; Twin Falls, ID; Uintah, UT; Umatilla, OR; Union, OR; Union, SD; Valdez-Cordova, AK; Valencia, NM; Valley, MT; Van Wert, OH; Ventura, CA; Victoria, TX; Wabash, IL; Walla Walla, WA; Waller, TX; Wallowa, OR; Walsh, ND; Walworth, SD; Wapello, IA; Ward, ND; Ward, TX; Warren, IA; Waseca, MN; Washington, IA; Washington, MN; Washington, OR; Washoe, NV; Watonwan, MN; Waukesha, WI; Wayne, IN; Webb, TX; Weber, UT; Weld, CO; Whiteside, IL; Whitley, IN; Whitman, WA; Wichita, TX; Willacy, TX; Williamson, TX; Wilson, TX; Winnebago, WI; Wise, TX; Woodbury, IA; Wrangell, AK; Wright, IA; Yamhill, OR; Yankton, SD; Yavapai, AZ; Yellowstone, MT; Yoakum, TX; Yolo, CA; Yuma, AZ

Table A.3 continued from previous page

Primary Dealer	Counties Covered
Citi	Alameda, CA; Alexandria City, VA; Anne Arundel, MD; Arlington, VA; Baltimore City, MD; Baltimore County, MD; Bayamon, PR; Bell, TX; Bergen, NJ; Bexar, TX; Brazos, TX; Bronx, NY; Broward, FL; Brown, TX; Bucks, PA; Burnet, TX; Caguas, PR; Camden, NJ; Carson City, NV; Chester, PA; Clark, NV; Clay, TX; Coleman, TX; Collin, TX; Comanche, TX; Contra Costa, CA; Cook, IL; Cooke, TX; Dallas, TX; Delaware, PA; Denton, TX; District of Columbia, DC; DuPage, IL; Duval, FL; Eastland, TX; Elko, NV; Ellis, TX; Erie, NY; Essex, MA; Essex, NJ; Fairfax, VA; Fairfield, CT; Falls, TX; Fort Bend, TX; Fresno, CA; Guadalupe, TX; Guam, GU; Harris, TX; Hartford, CT; Haskell, TX; Hudson, NJ; Humboldt, NV; Hunt, TX; Imperial, CA; Jack, TX; Johnson, TX; Kane, IL; Kankakee, IL; Kern, CA; Kings, CA; Kings, NY; Lake, IL; Lampasas, TX; Los Angeles, CA; Loudoun, VA; Lynn, TX; Lyon, NV; Madera, CA; Marin, CA; McHenry, IL; Merced, CA; Mercer, NJ; Miami-Dade, FL; Middlesex, MA; Middlesex, NJ; Midland, TX; Minnehaha, SD; Monmouth, NJ; Montague, TX; Monterey, CA; Montgomery, MD; Montgomery, PA; Montgomery, TX; Morris, NJ; Napa, CA; Nassau, NY; Nevada, CA; New Castle, DE; New Haven, CT; New York, NY; Norfolk, MA; Orange, CA; Palm Beach, FL; Palo Pinto, TX; Parker, TX; Passaic, NJ; Philadelphia, PA; Placer, CA; Plymouth, MA; Prince George's, MD; Prince William, VA; Queens, NY; Richmond, NY; Riverside, CA; Rockland, NY; Rockwall, TX; Runnels, TX; Sacramento, CA; San Bernardino, CA; San Diego, CA; San Francisco, CA; San Joaquin, CA; San Juan, PR; San Luis Obispo, CA; San Mateo, CA; San Saba, TX; Santa Barbara, CA; Santa Clara, CA; Santa Cruz, CA; Shackelford, TX; Solano, CA; Somerset, NJ; Sonoma, CA; Stanislaus, CA; Suffolk, MA; Suffolk, NY; Tarrant, TX; Taylor, TX; Tom Green, TX; Travis, TX; Tulare, CA; Union, NJ; Ventura, CA; Washoe, NV; Westchester, NY; Wichita, TX; Wilbarger, TX; Will, IL; Wise, TX; Young, TX

Table A.4: Physical Presence of Non-Bank U.S. Primary Dealers

This table identifies the counties where U.S. primary dealers without commercial bank subsidiaries maintained a physical presence as of 2008. Share weights are approximated using a composite of office square footage and employee headcounts disclosed in annual reports. The paper's key results are robust to alternative weighting schemes, provided the substantial share allocated to New York County is maintained.

Primary Dealer	Counties Covered
Morgan Stanley	Anchorage, AK; Jefferson, AL; Mobile, AL; Pulaski, AR; Maricopa, AZ; Pima, AZ; San Francisco, CA; San Mateo, CA; Santa Clara, CA; Alameda, CA; Contra Costa, CA; Sacramento, CA; Los Angeles, CA; Orange, CA; San Diego, CA; Ventura, CA; Santa Barbara, CA; Fresno, CA; Riverside, CA; San Bernardino, CA; Marin, CA; Denver, CO; Arapahoe, CO; El Paso, CO; Jefferson, CO; Fairfield, CT; Hartford, CT; New Haven, CT; District of Columbia, DC; New Castle, DE; Miami-Dade, FL; Palm Beach, FL; Broward, FL; Pinellas, FL; Hillsborough, FL; Orange, FL; Sarasota, FL; Collier, FL; Duval, FL; Alachua, FL; Leon, FL; Fulton, GA; DeKalb, GA; Cobb, GA; Gwinnett, GA; Honolulu, HI; Polk, IA; Ada, ID; Cook, IL; DuPage, IL; Lake, IL; Sangamon, IL; Marion, IN; Allen, IN; Johnson, KS; Jefferson, KY; Orleans, LA; East Baton Rouge, LA; Suffolk, MA; Middlesex, MA; Norfolk, MA; Worcester, MA; Essex, MA; Hampden, MA; Baltimore, MD; Baltimore City, MD; Montgomery, MD; Anne Arundel, MD; Prince George's, MD; Cumberland, ME; Wayne, MI; Oakland, MI; Kent, MI; Hennepin, MN; Ramsey, MN; St. Louis City, MO; Jackson, MO; Hinds, MS; Yellowstone, MT; Mecklenburg, NC; Wake, NC; Guilford, NC; Forsyth, NC; Cass, ND; Douglas, NE; Hillsborough, NH; Hudson, NJ; Essex, NJ; Morris, NJ; Bergen, NJ; Middlesex, NJ; Monmouth, NJ; Union, NJ; Mercer, NJ; Burlington, NJ; Camden, NJ; Bernalillo, NM; Clark, NV; Washoe, NV; New York, NY; Westchester, NY; Kings, NY; Queens, NY; Nassau, NY; Suffolk, NY; Rockland, NY; Albany, NY; Erie, NY; Monroe, NY; Onondaga, NY; Franklin, OH; Cuyahoga, OH; Hamilton, OH; Summit, OH; Montgomery, OH; Lucas, OH; Oklahoma, OK; Tulsa, OK; Multnomah, OR; Clackamas, OR; Lane, OR; Philadelphia, PA; Montgomery, PA; Delaware, PA; Chester, PA; Bucks, PA; Allegheny, PA; Lehigh, PA; York, PA; San Juan, PR; Providence, RI; Richland, SC; Greenville, SC; Minnehaha, SD; Davidson, TN; Shelby, TN; Knox, TN; Harris, TX; Dallas, TX; Tarrant, TX; Travis, TX; Bexar, TX; Collin, TX; Denton, TX; El Paso, TX; Salt Lake, UT; Davis, UT; Utah, UT; Fairfax, VA; Arlington, VA; Richmond City, VA; Virginia Beach City, VA; Chittenden, VT; King, WA; Spokane, WA; Pierce, WA; Snohomish, WA; Milwaukee, WI; Dane, WI; Kanawha, WV; Cabell, WV; Laramie, WY
Goldman Sachs	New York, NY; Albany County, NY; Los Angeles, CA; San Francisco, CA; San Mateo, CA; Fairfield, CT; District of Columbia, DC; Miami-Dade, FL; Palm Beach, FL; Fulton, GA; Cook, IL; Suffolk, MA; Worcester, MA; Mecklenburg, NC; Hudson, NJ; Philadelphia, PA; Dallas, TX; Harris, TX; Salt Lake, UT; Henrico, VA; King, WA; Denver, CO

Table A.4 continued from previous page

Primary Dealer	**Counties Covered**
Jefferies	New York County, NY; Hudson County, NJ; Essex County, NJ; Fairfield County, CT; Suffolk County, MA; Philadelphia County, PA; District of Columbia, DC; Richmond City, VA; Fulton County, GA; Davidson County, TN; Orleans Parish, LA; Cook County, IL; Dallas County, TX; Harris County, TX; Los Angeles County, CA; Santa Clara County, CA; San Francisco County, CA
Cantor Fitzgerald	New York County, NY; Bergen County, NJ; Monmouth County, NJ; Fairfield County, CT; Suffolk County, MA; Cook County, IL; Dallas County, TX; Los Angeles County, CA

Table A.5: First-Stage Instrument Relevance for County-Level Deposits

This table reports first-stage regression results at the county-year level. The dependent variable is the annual change in county-level deposits for columns (1)–(4). Both the dependent variable and the deposit channel instrument are scaled by the previous year's total county-level deposits and winsorized at the 1st and 99th percentiles. The Full Sample (2009–2023), QE I–III (2009–2014), Pandemic QE (2020–2021), and No QE (2015-2019, 2022–2023) periods are shown in columns (1) through (4), respectively. All specifications include county and year fixed effects. Robust standard errors, two-way clustered at the county and state-year levels, are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta\text{Deposits}_{c,t}/\text{Deposits}_{c,t-1}$			
$Z_{c,t}/\text{Deposits}_{c,t-1}$	0.42***	1.00***	1.68***	0.27*
	(0.05)	(0.15)	(0.26)	(0.14)
N	46,649	18,679	6,216	21,745
R^2	0.31	0.28	0.53	0.25
F-stat	66.40	44.78	40.79	3.78
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Sample Period	Full Sample	QE I–III	Pandemic QE	No QE

Table A.6: First-Stage Instrument Relevance for Bank-County-Level Deposits

This table reports first-stage regression results at the bank-county-year level. The dependent variable is the annual change in bank-county deposits. Both the dependent variable and the deposit channel instrument are scaled by the previous year's total bank-level assets and winsorized at the 1st and 99th percentiles. Column (1) covers the Full Sample period (2009–2023). Column (2) restricts the sample to small banks, defined as those with total assets below $1 billion as of year-end 2008. Columns (3) and (4) report results for the QE I–III (2009–2014) and Pandemic QE (2020–2021) sub-periods, respectively. All specifications include bank×year and county×year fixed effects; Columns (1) and (2) further include bank×county fixed effects. Robust standard errors, two-way clustered by bank and county, are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta\text{Deposits}_{b,c,t}/\text{Assets}_{b,t-1}$			
$Z_{b,c,t}/\text{Assets}_{b,t-1}$	0.91***	1.11***	0.31***	1.95***
	(0.07)	(0.09)	(0.06)	(0.08)
N	289,815	134,346	112,175	40,325
R^2	0.66	0.72	0.56	0.59
F-stat	161.81	161.77	25.67	542.65
Bank×Year FE	Yes	Yes	Yes	Yes
County×Year FE	Yes	Yes	Yes	Yes
Bank×County FE	Yes	Yes	No	No
Bank Size	All	Small	All	All
Sample Period	Full Sample	Full Sample	QE I–III	Pandemic QE

Table A.7: The Deposit Channel of QE: Effects on Bank-County Lending

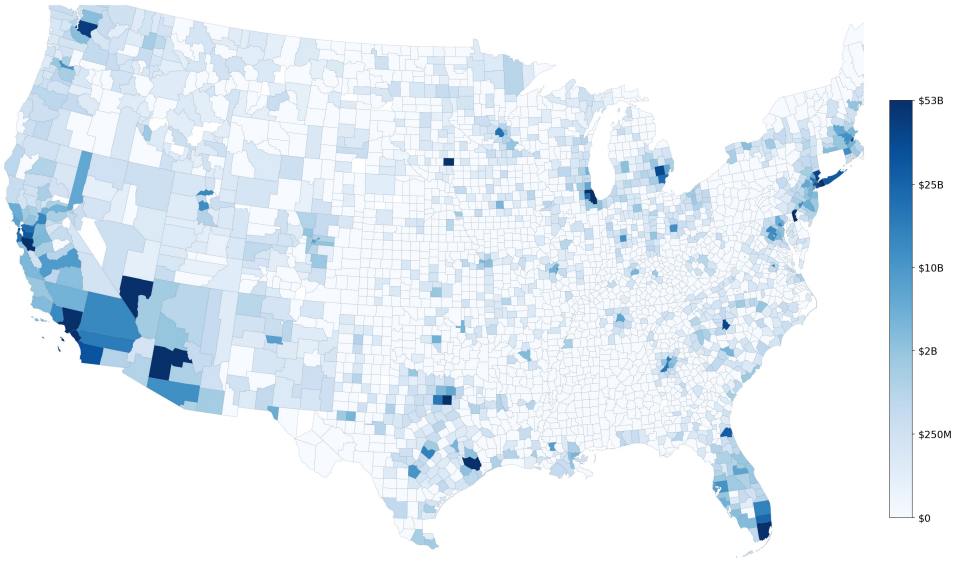
This table reports regression results examining the effect of the deposit channel of QE on bank-county-level lending outcomes for counties within Metropolitan Statistical Areas (MSAs). Column (1) presents the results for mortgage lending for the full sample. Column (2) reports mortgage lending for the subset of bank-counties that also engage in small business (SMB) lending. Column (3) reports small business lending for the full sample. Both the dependent variables and the deposit channel instrument are normalized by the previous year's total bank-level assets. The sample covers 2009–2023, with all variables winsorized at the 1st and 99th percentiles. All specifications include bank×year, county×year, and bank×county fixed effects, as well as an indicator for the presence of a physical bank branch in the county. Robust standard errors, two-way clustered at the bank and county levels, are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)
Dependent Variable:	$\Delta\text{Mortgages}_{b,c,t}/\text{Assets}_{b,t-1}$	$\Delta\text{SMB}_{b,c,t}/\text{Assets}_{b,t-1}$	
$Z_{b,c,t}/\text{Assets}_{b,t-1}$	1.57***	0.38***	0.40***
	(0.10)	(0.02)	(0.02)
In Branch	0.0006***	0.0003***	0.0003***
	(0.0001)	(0.0001)	(0.0001)
N	602,886	602,886	1,006,113
R^2	0.87	0.89	0.87
Bank×Year FE	Yes	Yes	Yes
County×Year FE	Yes	Yes	Yes
Bank×County FE	Yes	Yes	Yes
Sample Period	Full Sample	Full Sample	Full Sample

Figure A.1: Geographic Distribution of Predicted and Actual Deposit Growth

This figure presents county-level maps comparing predicted deposit inflows transmitted through primary-dealer networks with actual deposit growth across U.S. counties over 2009–2023. Panel A maps cumulative predicted deposit inflows by county, constructed from the county-level deposit channel instrument, and Panel B maps the actual change in total county-level deposits over the same period. In both panels, darker shading indicates larger dollar amounts. The two distributions closely correspond: counties with greater exposure to primary-dealer activity exhibit both larger predicted inflows and greater realized deposit growth, consistent with the first-stage relevance of the instrument.

Panel A: Predicted Deposit Inflows



Panel B: Actual Deposit Change

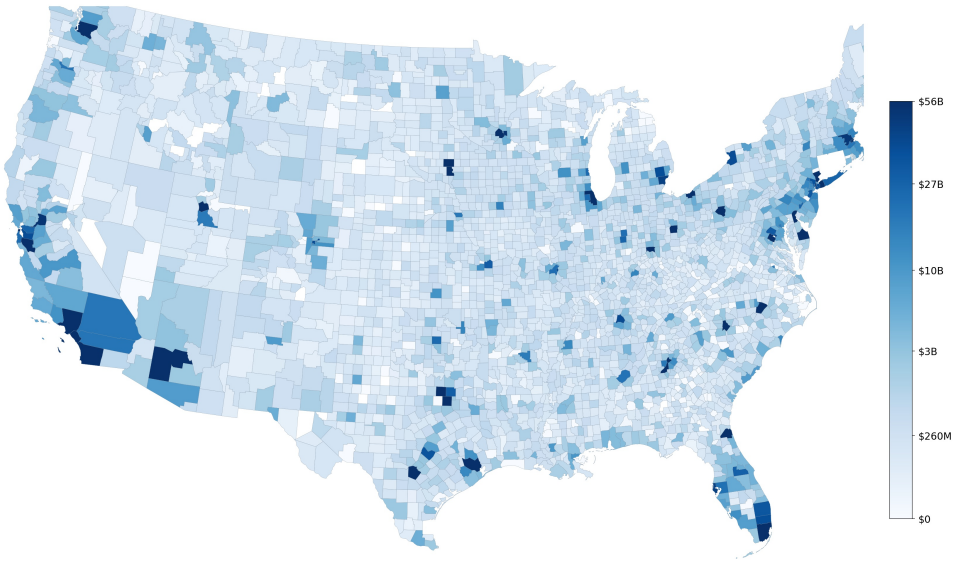
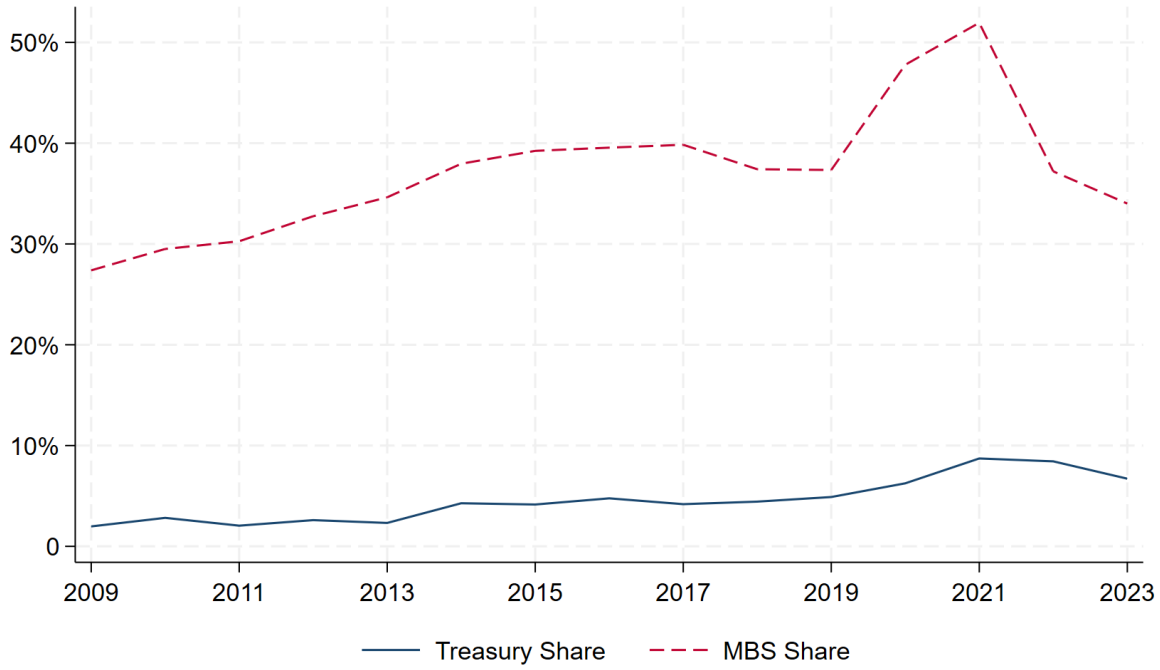


Figure A.2: U.S. Bank Ownership of Publicly Held Treasuries and MBS Securities

This figure plots the time-series of total U.S. Treasuries and Mortgage-Backed Securities (MBS) held by the commercial banking sector as a percentage of the total volume outstanding with the public. Treasuries held by the public are defined as total financial system holdings (BOGZ1FL893061105Q) net of Federal Reserve holdings (BOGZ1FL713061103Q). Publicly held MBS is calculated as the sum of government-sponsored entity (GSE) (BOGZ1FL403065005Q) and non-GSE holdings (AGSEBMPTCMAHDFS), less MBS held by the Federal Reserve (BOGZ1FL713061903Q).



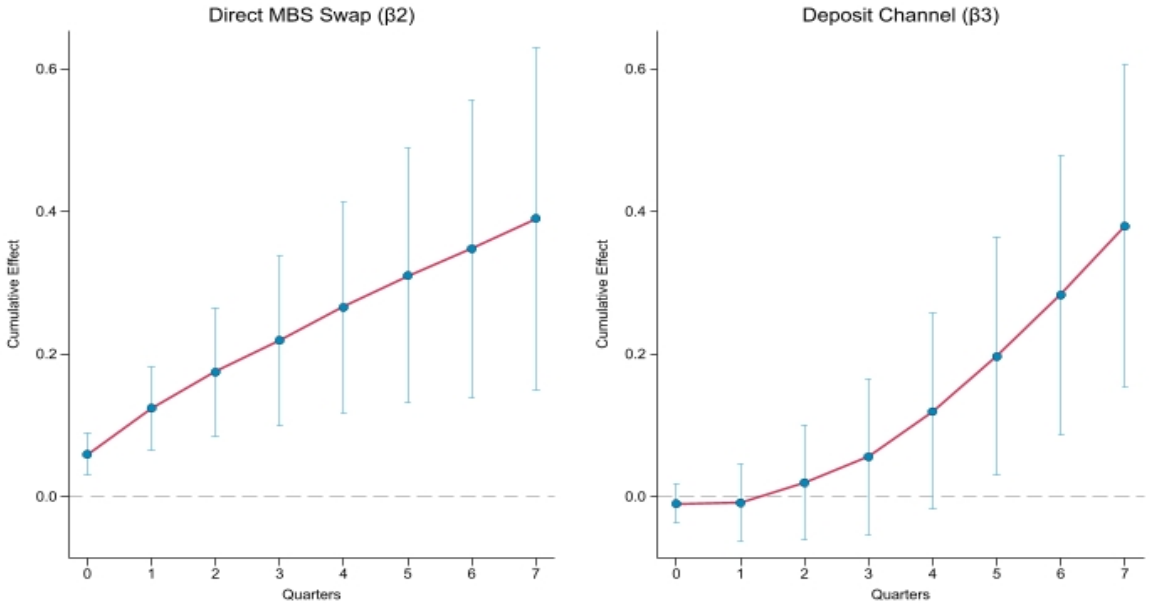
Source: Federal Reserve Economic Data (FRED), series BOGZ1FL893061105Q, BOGZ1FL713061103Q, AGSEBMPTCMAHDFS, BOGZ1FL403065005Q, BOGZ1FL713061903Q, BOGZ1LM763061100Q, and BOGZ1LM763061705Q.

Figure A.3: Local Projections of the Direct MBS-Swap and Deposit Channel Effects on Loan Categories

This figure plots the impulse responses of future bank mortgage and C&I lending to the direct MBS-swap effect and the deposit channel of QE. Panels A and B display the results for mortgage loans and C&I loans, respectively. Detailed variable definitions are provided in Table 6. Error bars represent 95% confidence intervals.

$$(Y_{b,t+h} - Y_{b,t-1})/\text{Assets}_{b,t-1} = \beta_{1h} Z_{\text{tsy},b,t}/\text{Assets}_{b,t-1} + \beta_{2h} Z_{\text{mbs},b,t}/\text{Assets}_{b,t-1} + \beta_{3h} Z_{b,t}/\text{Assets}_{b,t-1} + \alpha_b + \gamma_t + \epsilon_{b,t}, \qquad h = 0, 1, \cdots, 7.$$

Panel A: Mortgage Loans



Panel B: C&I Loans

